

A Novel Climate-Change Fingerprint Framework for Quantile-Dependent Thermal Extremes and Compound Dry-Hot Signal Emergence Across Iran

Running title: Climate-change fingerprints across Iran

Abstract

Mean temperature alone cannot fully describe thermal extremes. Many impacts arise from changes in distributional shape, tail behavior, and the concurrence of heat with moisture deficit, not just from shifts in the center of the temperature distribution. A reproducible observational climate-change fingerprint framework was developed for examining quantile-dependent trends in four annual thermal-extreme indices across Iran during 1991-2024 using observations from 124 meteorological stations. In this framework, fingerprint refers to cross-diagnostic consistency among warming-related signals, not to formal external-forcing attribution. Warm days, warm nights, cool days, and cool nights were evaluated with station-wise quantile regression and moving-block bootstrap uncertainty estimation. The workflow also included false-discovery-rate field-significance screening, spatial diagnostics, cluster-based regionalization, a Quantile Climate-Fingerprint and Emergence Analysis (QCF-EA), Köppen-Geiger climate-regime stratification, and an empirical-copula analysis of compound dry-hot years based on precipitation and temperature. The results point to a consistent but spatially uneven warming pattern. Warm days exhibit the strongest regional increase and the most pronounced upper-tail amplification. Regional network-mean slopes are $+9.45$, $+15.55$, and $+22.50$ days per decade at $q0.10$, $q0.50$, and $q0.90$, respectively, with $\Delta1 = +13.06$. Warm nights also increase, but more evenly across the distribution ($+8.74$, $+12.14$, $+12.21$; $\Delta1 = +3.47$). Cool days and cool nights decline. The contraction is stronger for cool days at the upper quantile (-5.60 , -12.00 , -15.37 ; $\Delta1 = -9.77$) than for cool nights (-5.91 , -9.83 , -12.06 ; $\Delta1 = -6.16$). At $q0.50$, FDR screening retained 115/124 warm-day, 104/124 warm-night, 97/124 cool-day, and 86/124 cool-night station results. The QCF-EA layer gives the climate-change interpretation an additional observational check. The regional station-temperature anomaly increased by $+0.446$ deg C per decade; fixed-baseline late-period shifts were $+36.78$ warm days, $+28.41$ warm nights, -10.69 cool days, and -8.19 cool nights per year relative to 1991-2007; and $q0.90$ responses to regional warming were $+29.62$, $+22.60$, -21.03 , and -19.66 days per year per deg C for the same four indices. Köppen-Geiger stratification shows a national warming pattern with clear regime-level differences. Fingerprint scores range from 0.847 in cold-desert stations to 0.925 in cold dry-summer stations. The compound layer indicates that rare dry-hot station-year concurrence has also become more widespread, especially in the warm season: $RP \geq 10$ dry-hot years affected an average of 10.35 stations in 1991-2007 and 41.53 stations in 2008-2024 (Kendall tau = 0.583 , $p < 0.001$), while $RP \geq 20$ events increased from 2.59 to 16.12 stations. The dominant driver of warm-season $RP \geq 10$ dry-hot events shifted from dryness in the early period (147 dry-dominant versus 27 hot-dominant station-events) to heat in the late period (141 versus 535). The integrated climate-fingerprint score was 0.897 , while the network quantile-fingerprint score was 1.000 under a circular-shift null ($p = 0.002$). Overall, the evidence indicates an observational pattern of recent warming-related restructuring in thermal extremes across Iran, with a complementary dry-hot signal that links the temperature results more directly to hydroclimatic risk, while formal anthropogenic attribution remains beyond the scope of the study.

Keywords: thermal extremes; quantile regression; compound dry-hot extremes; climate fingerprint; climate change; warm days; cool days; regionalization; Iran

1. Introduction

Climate change is often experienced most directly through extremes, not averages, because extremes are what disrupt agriculture, water resources, ecosystems, infrastructure, and human health (Easterling et al., 2000; IPCC, 2021). Recent work shows that hot extremes have continued to increase despite internal variability, that a large share of high-temperature extremes is attributable to warming, and that record-shattering extremes become more likely as warming proceeds (Coumou and Rahmstorf, 2012; Seneviratne et al., 2014; Fischer and Knutti, 2015; Fischer et al., 2021). Heat-related risk also depends on exposure and physiological vulnerability, with deadly-heat exposure and heat-attributable mortality already detectable in multi-region assessments (Mora et al., 2017; Vicedo-Cabrera et al., 2021; Peng et al., 2026; Perkins-Kirkpatrick et al., 2026). Heat-related risk, in particular, depends strongly on what happens in the tails of the thermal distribution, not on the mean alone (Katz and Brown, 1992; Katz et al., 2002; Perkins, 2015). A region may therefore show only modest mean change while still undergoing marked upper-tail intensification or a rapid loss of historically cool conditions. Distribution-aware methods are especially valuable when the aim is to understand how thermal extremes are changing (Koenker and Bassett, 1978; Koenker, 2005; Barbosa et al., 2011; Fan, 2014; Reich, 2012).

That perspective is particularly relevant for annual warm-day, warm-night, cool-day, and cool-night indices. These measures are widely used in climatology because they translate daily temperature variability into annual counts that are easy to interpret physically (Frich et al., 2002; Alexander et al., 2006; Zhang et al., 2011; Dunn et al., 2020). Even so, many studies still concentrate on mean or median behavior, despite the fact that rates of change can differ substantially between lower, central, and upper quantiles. Those differences matter because they help separate broad, distribution-wide shifts from more selective amplification in the most extreme years (Moberg and Jones, 2005; Kostopoulou and Jones, 2005; Fischer and Schär, 2010; Russo and Sterl, 2011). Recent work on heatwave dynamics and land-atmosphere feedbacks also shows that the upper tail can respond differently from the mean when circulation persistence, aridity, evapotranspiration, or soil moisture-temperature coupling changes (McKinnon et al., 2021; Rousi et al., 2022; Vautard et al., 2023; Maraun et al., 2025; García-García et al., 2023).

Thermal risk also becomes more consequential when heat coincides with moisture deficit. Compound dry-hot events can produce impacts that exceed those expected from either driver alone, because heat can intensify evaporative demand while low precipitation can reinforce land-surface drying and sensible-heat partitioning (Leonard et al., 2014; Zscheischler and Seneviratne, 2017; Zscheischler et al., 2018). Recent empirical and probabilistic studies therefore increasingly treat heat and dryness as jointly distributed hazards rather than as separate univariate extremes (Hao et al., 2013; Sarhadi et al., 2018; Alizadeh et al., 2020). For a dryland and water-stressed setting such as Iran, this compound framing is not a side issue. It provides a way to ask whether the warming signal identified in temperature indices is also accompanied by years in which heat and precipitation deficit occur together across a larger share of the station network.

Iran offers a particularly informative setting in which to examine this question. The country includes humid Caspian lowlands, the Zagros and Alborz mountain systems, central desert basins, and hot southern coasts influenced by the Persian Gulf and Gulf of Oman. Such strong contrasts in elevation, continentality, aridity, and marine influence are likely to produce spatially differentiated responses rather than a single national thermal signal. Previous studies have already documented intensifying temperature extremes across Iran and the broader Middle East and North Africa (Zhang et al., 2005a; Soltani et al., 2016; Vaghefi et al., 2019; Zittis et al., 2016; Francis and Fonseca, 2024; Naderi et al., 2024; Jahanshahi et al., 2024), and regional work has drawn

increasing attention to heat stress and compound thermal hazards in southwest Asia (Pal and Eltahir, 2016; Raymond et al., 2020; Zittis et al., 2021; Raymond et al., 2024). Recent station-based Köppen-Geiger analysis also indicates observed warming-driven aridification and climate-type transitions across Iran (Jamali et al., 2026). What remains less common, however, is a dense station-based assessment for Iran that brings together distributional trend analysis, uncertainty quantification, field significance, and quantitative regionalization in a single study while also distinguishing clearly between distribution-relative diagnostics and fixed-baseline sensitivity checks.

To address that gap, we use a denser contemporary station network than is typical of national syntheses based on long records. In this manuscript, the term climate-change fingerprint framework denotes a structured observational workflow for testing whether multiple diagnostics point in the same warming-consistent direction; it does not imply a formal counterfactual attribution design. Using 124 meteorological stations for 1991–2024, we analyze annual warm days, warm nights, cool days, and cool nights with quantile regression, bootstrap uncertainty estimation, field-significance screening, spatial diagnostics, and cluster-based regionalization. We then add an empirical-copula dry-hot module that uses station precipitation and temperature jointly to evaluate whether warming-related thermal change is accompanied by more widespread compound heat-moisture stress. In the title, compound dry-hot signal emergence therefore refers to the increasing empirical prevalence, spatial connectedness, and heat-dominant character of dry-hot station-years within the observed archive, not to a formal detection-and-attribution threshold. Our goal is not simply to ask whether thermal extremes are changing, but to examine how those changes are distributed across the response surface, how they organize in space, how stable they remain under alternative methodological choices, and whether their hydroclimatic context is also changing. This kind of physically grounded evidence is relevant to climate-risk assessment, because heat-risk planning increasingly depends on knowing which parts of the thermal distribution are changing, where those changes are emerging, and which climatic settings may require more targeted monitoring (Mora et al., 2017; Vicedo-Cabrera et al., 2021; Fischer et al., 2021).

The novelty of the approach lies in the reproducible integration of several complementary diagnostics rather than in any single statistical tool. The analysis makes eight related contributions that move the study beyond a descriptive inventory of station trends. It resolves change across the conditional distribution instead of relying on mean behavior alone; measures asymmetry with explicit tail-contrast metrics; quantifies uncertainty through dependent bootstrap resampling and complementary significance tests; regionalizes station behavior through reproducible clustering and representative-station selection; tests the stability of the main findings through homogeneity-exclusion, bootstrap-depth, bootstrap-method, interpolation, clustering, and split-period analyses; adds a climate-fingerprint layer that is treated as observational signal detection, not formal attribution; evaluates how those patterns differ across Köppen-Geiger climate regimes; and adds a compound dry-hot analysis that links the thermal signal to precipitation deficit using empirical joint probabilities. The resulting framework is novel primarily as an integrated, reproducible synthesis for Iran: it combines distributional thermal diagnostics, emergence-style consistency checks, climate-regime stratification, and compound dry-hot concurrence in one station-based workflow. These components provide a distribution-aware and hydroclimatically contextualized synthesis of recent thermal-extreme change across Iran.

2. Data and methods

2.1. Data and study region

The analysis is based on daily station temperature and precipitation observations from 124 meteorological stations distributed across Iran for the common period 1991–2024. Temperature observations were used to construct the four thermal-extreme indices, while precipitation observations were used in the compound dry-

hot module. After preprocessing, each station-index series contributes 34 annual values. The network spans coastal, lowland, plateau, and high-elevation settings, and it is dense enough to capture substantial physiographic heterogeneity while remaining entirely observation-based.

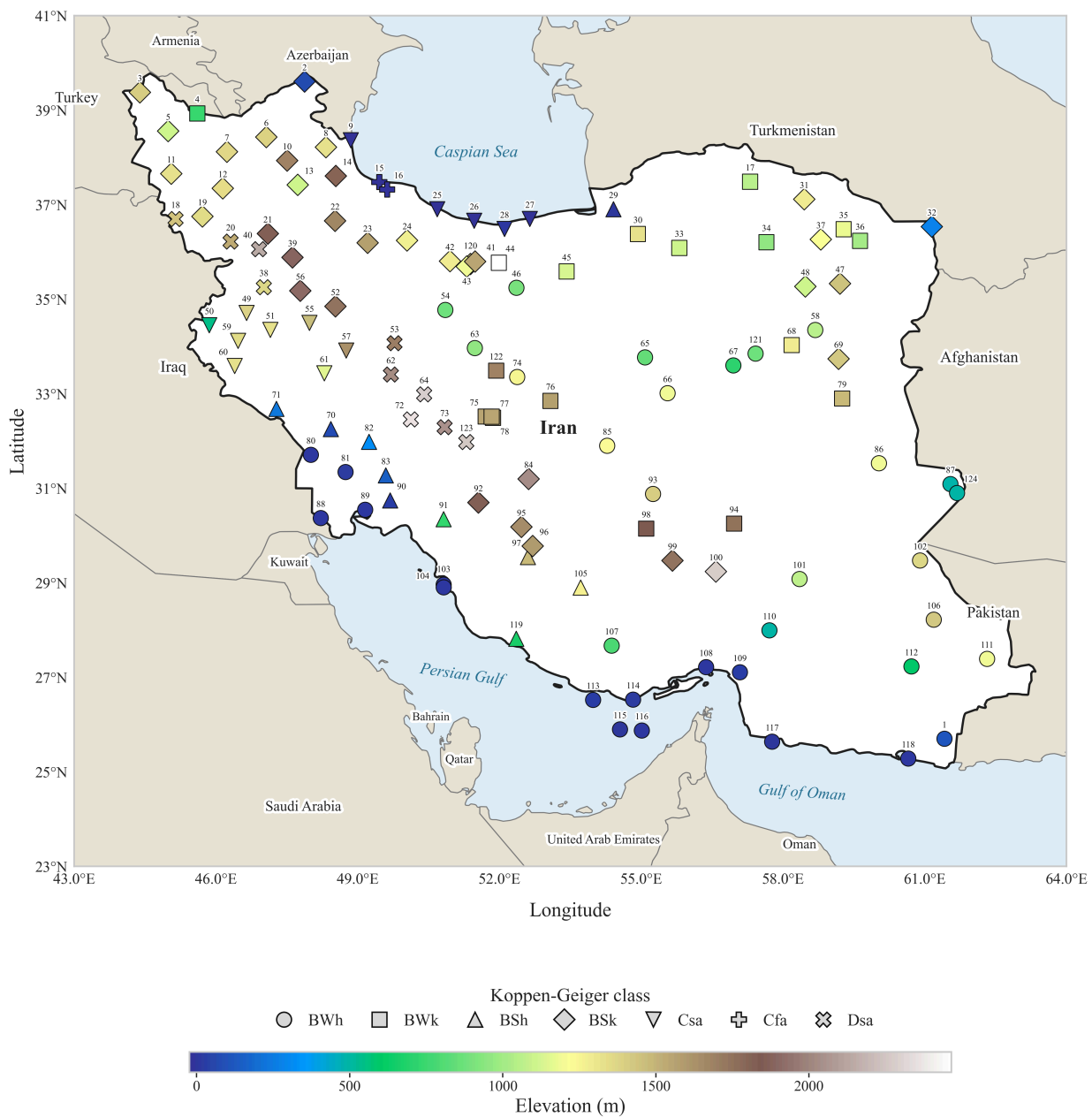


Figure 1. Regional context and station network used in the analysis. Stations are colored by elevation and numbered by ascending `station_id`; neighboring countries and the Caspian Sea, Persian Gulf, and Gulf of Oman are shown to place the Iranian station network in its wider geographical setting.

The choice to work at the station level is deliberate. All primary inference is anchored in observed station records rather than interpolated grid cells, which is especially important in a region where elevation contrasts and marine influence can produce sharp local thermal differences over short distances. The daily archive includes minimum, maximum, and, where available, mean temperature, together with daily precipitation totals. Leap-day observations were removed before percentile-threshold estimation so that the annual cycle remained consistent at 365 days.

Data-quality screening showed generally high completeness across the network, with median station completeness of 98.96% for `tmin`, 99.42% for `tmax`, and 99.35% for `tmean`; daily precipitation completeness was also high (98.94% across the archive), which made a station-based compound dry-hot sensitivity layer

feasible. No station contained duplicate dates. Even so, internal consistency checks identified at least one $t_{min} > t_{max}$ conflict at 32 stations and at least one t_{mean} value outside the daily temperature range at 78 stations, confirming that quality control remained necessary despite the high overall coverage. Homogeneity tests applied to annual mean temperature were highly sensitive in raw form, flagging 119/124 stations, whereas detrended versions still flagged 39/124 stations. This contrast suggests that breakpoint-sensitive homogeneity statistics respond strongly in a region experiencing pervasive warming. They should not be interpreted mechanically as direct evidence requiring correction of the daily series.

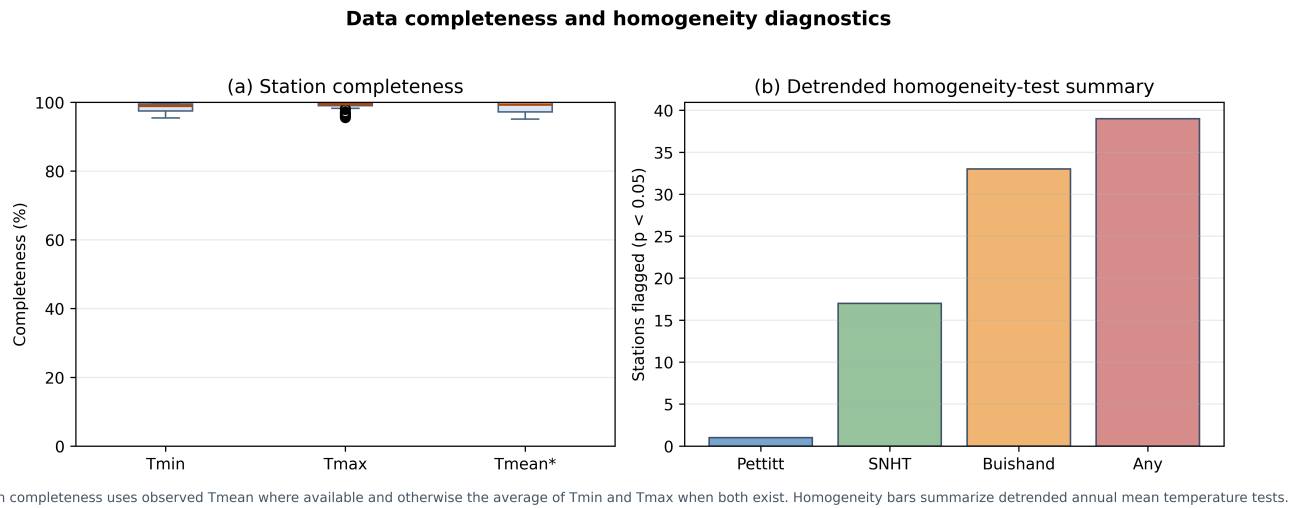


Figure 2. Combined overview of station-level completeness and detrended homogeneity diagnostics for the 1991-2024 station archive. The left panel summarizes completeness, and the right panel summarizes the number of stations flagged by detrended homogeneity tests.

No automatic homogenization adjustment was imposed on the daily temperature records. That decision reflects a practical constraint: the available archive does not include the station-history metadata or neighbor-based reference framework needed for a defensible daily homogenization procedure, and the main breakpoint tests were applied to annual mean temperature instead of directly to the percentile-based daily series used for the indices (Zhang et al., 2005b). We therefore retain the original daily observations, treat the tests as screening information, and use flagged-station exclusion as a conservative sensitivity bound rather than as a substitute for full homogenization. Supporting summaries are provided in Supplementary Figure S1 and Supplementary Tables S1-S3, while the exclusion sensitivity is reported in Supplementary Figure S8 and Supplementary Table S4.

2.2. Thermal-extreme indices

Four annual thermal-extreme indices were constructed from daily minimum (t_{min}) and maximum (t_{max}) temperature: warm days, warm nights, cool days, and cool nights. For each station, day-of-year-specific thresholds were estimated from the available 1991-2024 record using a 5-day moving window and the 10th and 90th percentiles. We required a minimum of 15 reference samples per day-of-year window; when local window support fell below that threshold, the station-wide reference sample was used as a fallback.

Warm days and cool days were defined from daily maximum temperature relative to the station-specific 90th and 10th percentile thresholds, respectively. Warm nights and cool nights were defined in the same way from daily minimum temperature. Binary exceedance indicators were then summed by year to obtain annual counts (Frich et al., 2002; Alexander et al., 2006; Zhang et al., 2011). Years with less than 80% valid daily coverage for the corresponding temperature variable were treated as missing at the annual-index stage.

Because only 1991–2024 data were available, the threshold climatology is based on the full analysis period rather than on a separate earlier reference interval. The resulting indices therefore describe how annual frequencies of warm and cool events evolved relative to the empirical distribution of the period being analyzed. This choice yields internally consistent station-wise indices for the available dataset, but it also means that annual counts should be read as distribution-relative measures, not as departures from a fixed pre-change climatological baseline. Put simply, the main index set is designed to detect redistribution within the observed record, while the fixed-baseline checks test whether the same directional story survives when an earlier reference window is imposed.

2.3. Quantile regression and summary metrics

Quantile regression was used as the main inferential tool because our interest lies in distribution-dependent change rather than mean behavior alone (Koenker and Bassett, 1978; Koenker, 2005). For an annual response Y and time covariate T , the τ -level conditional quantile is written as

$$Q_Y(\tau | T) = \beta_0(\tau) + \beta_1(\tau)T, \quad \tau \in (0, 1).$$

The estimator is obtained from the asymmetric absolute-loss objective

$$\hat{\beta}(\tau) = \arg \min_{\beta} \sum_{i=1}^n \rho_{\tau}(y_i - x_i^{\top} \beta),$$

where $x_i = (1, T_i)^{\top}$ and

$$\rho_{\tau}(u) = u(\tau - \mathbf{1}\{u < 0\}).$$

This framework permits slope heterogeneity across the distribution, so $\beta_1(\tau)$ can differ meaningfully between lower, central, and upper quantiles. Time was expressed in decades, making all slopes interpretable as changes in annual event counts per decade.

The full quantile profile was estimated from $q = 0.10$ to $q = 0.90$ at increments of 0.01 . In the main text, interpretation centers on $q0.10$, $q0.50$, and $q0.90$, together with an OLS benchmark slope. In addition to station-wise fits, we also derived descriptive regional quantile profiles by averaging each annual index across all stations and fitting quantile regression to the resulting network-mean time series. These regional profiles are intended for synthesis and visualization, whereas the main inferential backbone remains the station-level analysis.

To summarize distributional asymmetry, three contrasts were computed:

$$\Delta_1 = \text{slope}(q0.90) - \text{slope}(q0.10)$$

$$\Delta_2 = \text{slope}(q0.90) - \text{slope}(q0.50)$$

$$\Delta_3 = \text{slope}(q0.50) - \text{slope}(q0.10)$$

Positive Δ_1 indicates stronger upper-tail amplification, whereas negative Δ_1 indicates relative upper-tail weakening or, for declining indices, stronger upper-tail contraction.

2.4. Uncertainty, spatial significance, and regionalization

Bootstrap resampling was used to characterize uncertainty in the station-level quantile slopes. The baseline analysis used a moving-block bootstrap with 200 replicates. Because annual extreme-index series may retain

short-range temporal dependence, simple iid resampling was not adopted as the default. Block length was selected automatically with an $n^{(1/3)}$ rule and then bounded to the interval [2, 8], following the general dependent-bootstrap logic discussed by Kunsch (1989), Hall et al. (1995), and Lahiri (2003). For each focal quantile and for **Delta1**, we retained bootstrap means, standard deviations, and confidence intervals.

Inference in the tails relied primarily on the bootstrap summaries. This choice is important because the analytic tail outputs were not sufficient to support a stable national FDR screening layer for the available dataset, whereas bootstrap intervals remained available for the focal tail summaries. Tail-significance counts are therefore discussed on the basis of the bootstrap results, while field-significance screening is reported for the median quantile, where the analytic station-level outputs were complete.

Field significance was assessed with false-discovery-rate control (Benjamini and Hochberg, 1995), and spatial dependence was examined with Moran's I (Moran, 1950) using 499 permutations and 5-nearest-neighbor weights. In this setting, FDR controls the expected proportion of false positives among the station-level results declared significant after multiple testing. That distinction is important because the study evaluates many station-wise trend tests in parallel; without multiplicity control, some apparently significant results would be expected simply by chance. FDR was preferred because it offers a practical balance between error control and statistical power, making it well suited to a dense multi-station setting in which an overly strict family-wise correction could suppress spatially meaningful signals. Regionalization was performed with hierarchical clustering using average linkage and Euclidean distance. The clustering stage was treated as exploratory but reproducible: it was based on a screened feature set derived from focal quantile slopes, delta contrasts, and selected bootstrap summaries, and it was accompanied by representative-station extraction, reduced-feature reruns, alternative clustering specifications, and a permutation-based spatial compactness test.

2.5. Additional synthesis analyses

Several additional analyses were used to assess the stability and interpretability of the main results. A homogeneity-exclusion sensitivity analysis repeated the regional summaries after excluding all stations flagged by any detrended homogeneity diagnostic. A bootstrap-depth sensitivity analysis repeated the bootstrap stage with 400 rather than 200 replicates. A bootstrap-method comparison contrasted the baseline moving-block results against maximum-entropy bootstrap (**meboot**) summaries (Efron and Tibshirani, 1993; Vinod, 2006). Interpolation sensitivity compared alternative spatial-display methods so that mapped surfaces could be interpreted as visualization products rather than hidden inferential layers. Alternative clustering solutions were used to examine sensitivity to linkage, distance metric, and algorithmic family. Split-period comparisons contrasted the first and second halves of the 1991-2024 record, namely **1991-2007** and **2008-2024**, using regional network-mean annual series. A compound dry-hot analysis used precipitation and temperature jointly to test whether thermal change was accompanied by more widespread heat-moisture concurrence. Finally, a simple driver analysis related slope and **Delta1** metrics to latitude, longitude, and elevation.

2.6. Quantile climate-fingerprint and emergence analysis

To strengthen the climate-change interpretation without overextending the study into formal attribution, we added a Quantile Climate-Fingerprint and Emergence Analysis (QCF-EA). This layer is the core fingerprint component of the framework named in the title. It was designed to test whether the observed station trends form a consistent warming-related pattern across indices, quantiles, uncertainty estimates, and independent baseline choices. It is used here as observational signal detection, not as a counterfactual attribution

experiment, and it is conceptually aligned with recent work on the emergence of climate-change signals in extreme-index distributions (Schuhen et al., 2026).

The QCF-EA has four components. First, the annual extreme indices were recomputed with fixed station-specific percentile thresholds derived only from 1991–2007. These thresholds were then applied to the full 1991–2024 record, and the late period (2008–2024) was compared with the fixed baseline period. This test addresses a limitation of the main distribution-relative indices by asking whether warm extremes become more frequent, and cool extremes less frequent, when the threshold definition is held fixed from the earlier half of the record.

Second, a regional station-temperature anomaly was constructed from annual station-mean temperature. For each station, annual mean temperature was expressed relative to its own 1991–2007 mean; the regional anomaly was then obtained by averaging station anomalies within each year. This gives an observation-based regional warming covariate. We then fitted quantile-response models of the form

$$Q_Y(\tau | A) = \alpha_0(\tau) + \alpha_1(\tau)A,$$

where A is the regional temperature anomaly in degrees Celsius. The coefficient $\alpha_1(\tau)$ is reported as days per year per degree Celsius and is interpreted as a warming-linked association rather than as an independent causal estimate, because both the temperature anomaly and the extreme indices are derived from the station archive.

Third, station-level signal emergence was summarized using bootstrap signal-to-noise ratios:

$$\text{SNR} = |\text{bootstrap mean slope}| / \text{bootstrap standard deviation}.$$

A station was classified as emerged for a given metric when the signal had the warming-consistent sign and $\text{SNR} \geq 2$. For warm indices, the expected sign is positive for both slopes and $\Delta 1$; for cool indices, the expected sign is negative. This rule is deliberately conservative in sign but descriptive in threshold choice, so it is a pragmatic emergence screen, not a universal physical threshold.

Fourth, the integrated climate-fingerprint score was computed as the mean of component scores representing median-trend direction, tail-asymmetry direction, fixed-baseline period shifts, warming-linked $q0.50$ responses, $q0.50$ signal emergence, FDR-retained median field significance, homogeneity-exclusion sign stability, and bootstrap-depth stability. This composite was intended as a transparent summary of concordance across complementary diagnostics rather than as a latent-variable model, so its value lies in the consistency of the underlying components rather than in any single numerical threshold. A complementary network fingerprint used the signs of the network-mean $q0.10$, $q0.50$, $q0.90$, and $\Delta 1$ responses across all four indices. Its null distribution was estimated with 499 circular-shift permutations of the annual network-mean series, which preserves the internal sequence of each index while disrupting its alignment with calendar year. This test evaluates whether the observed cross-index quantile pattern is unusually coherent relative to a temporally shifted null.

2.7. Köppen-Geiger climate-regime stratification

To test whether the national signal is uniform or climate-regime dependent, station locations were stratified using the present-day Köppen-Geiger classification of Beck et al. (2018). We sampled the 0.0083 deg WGS84 GeoTIFF map, representing 1980–2016 climate classes, at each station coordinate and retained the associated confidence value from the companion confidence raster. One coastal station falling on a water/no-data pixel was assigned to the nearest valid land pixel within the raster neighborhood. The resulting station labels were

then grouped into six analysis regimes with sufficient sample support: hot desert (BWh, $n = 36$), cold desert (BWk, $n = 18$), hot steppe (BSh, $n = 10$), cold steppe (BSk, $n = 35$), temperate (Csa/Cfa, $n = 15$), and cold dry-summer (Dsa, $n = 10$). The two Cfa stations were combined with the Csa stations because their separate sample size was too small for stable regime-level inference.

For each regime, we summarized station-level quantile slopes, Delta1, fixed-baseline period shifts, warming-linked quantile responses, and signal-emergence counts. We also computed a regime-level fingerprint score using the same component logic as QCF-EA, but restricted to stations in the corresponding climate regime. Between-regime differences were evaluated with 999 station-label permutation tests applied to regime-mean contrasts, followed by FDR correction across the tested metrics. These tests are used only to identify unusually large regime contrasts; they do not imply that the Köppen-Geiger class itself causally determines the observed trend.

2.8. Compound dry-hot empirical-copula analysis

To connect the thermal-extreme results to a hydroclimatic risk context, we added a compound dry-hot analysis inspired by empirical copula and multivariate return-period studies of concurrent dryness and heat (Corbella and Stretch, 2012; Sarhadi et al., 2018; Alizadeh et al., 2020). Within the broader fingerprint framework, this module tests whether the thermal signal has a hydroclimatic counterpart in years when heat and precipitation deficit occur together. The analysis was deliberately framed as a sensitivity and synthesis layer rather than as a formal drought-impact assessment. Precipitation deficit is a meteorological driver of dry conditions, whereas drought is an impact-oriented state that may also depend on soil moisture, water demand, management, and human modification (Van Loon et al., 2016). We therefore use the term dry-hot rather than drought-heatwave.

Two station-year definitions were evaluated. The annual definition paired total annual precipitation with annual mean temperature. The warm-season definition paired June-September precipitation with June-September mean daily maximum temperature, because this season is most directly linked to hot-season evaporative demand and heat stress in Iran. For each definition, station-year aggregates were retained only when both precipitation and temperature coverage were at least 80%. At least 25 valid station-years were required for empirical joint-probability estimation; 123/124 stations met the practical precipitation-support requirement for 30 or more valid annual and warm-season years.

For each station and definition, empirical joint probabilities were estimated without fitting a parametric marginal distribution or copula family. For a station-year with precipitation P_y and temperature T_y , the dry-hot AND probability was estimated as

$$\hat{p}_{AND,y} = \frac{1}{n} \sum_{i=1}^n \mathbf{1}(P_i \leq P_y, T_i \geq T_y),$$

and the empirical joint return period was

$$\widehat{RP}_{AND,y} = \frac{1}{\hat{p}_{AND,y}}.$$

This definition identifies years that are simultaneously dry relative to the station's own empirical precipitation distribution and hot relative to its own empirical temperature distribution. Because the record length is only 34 years, we did not use the 50- or 75-year return-period thresholds common in century-length compound-

event studies. Instead, we summarized $RP \geq 5$, $RP \geq 10$, and $RP \geq 20$ station-year events, with interpretation centered on $RP \geq 10$ and $RP \geq 20$.

For each year, we counted the number and percentage of valid stations experiencing a dry-hot event. Temporal change in the affected-station count was assessed with Kendall's rank trend test and a linear slope reported in stations per decade (Mann, 1945; Kendall, 1938). Early-late distributional shifts between 1991–2007 and 2008–2024 were tested with two-sample Kolmogorov-Smirnov, Cramer-von Mises, and Anderson-Darling empirical-distribution tests (Anderson and Darling, 1954; Massey, 1951). To examine whether compound events were becoming more spatially connected, annual binary event fields were evaluated with Moran's I using 5-nearest-neighbor weights and 499 permutations. Finally, the dominant station-event driver was classified by comparing the marginal empirical return periods of dry precipitation and hot temperature: events with larger dry return periods were labeled dry-dominant, events with larger hot return periods were labeled hot-dominant, and exact ties were treated as co-dominant. This driver label is descriptive; it does not establish causal ordering.

All compound dry-hot outputs were written to a separate project folder, `outputs/compound_dry_hot/`, with tables and figures stored under dedicated `tables/` and `figures/` subdirectories.

2.9. Analytical workflow

All analyses were implemented in a unified workflow that integrates data screening, annual-index construction, station-wise quantile regression, bootstrap uncertainty estimation, feature engineering, clustering, spatial diagnostics, climate-fingerprint diagnostics, climate-regime stratification, compound dry-hot analysis, and publication-oriented sensitivity modules. This workflow is the operational form of the climate-change fingerprint framework used in the manuscript: it asks whether independent observational diagnostics tell a coherent, warming-consistent story while preserving explicit uncertainty and scope limits. Bringing these steps together in one workflow reduces manual transfer error between stages and ensures that all tables, figures, and regional summaries used in this study are traceable to the same underlying run configuration.

3. Results and discussion

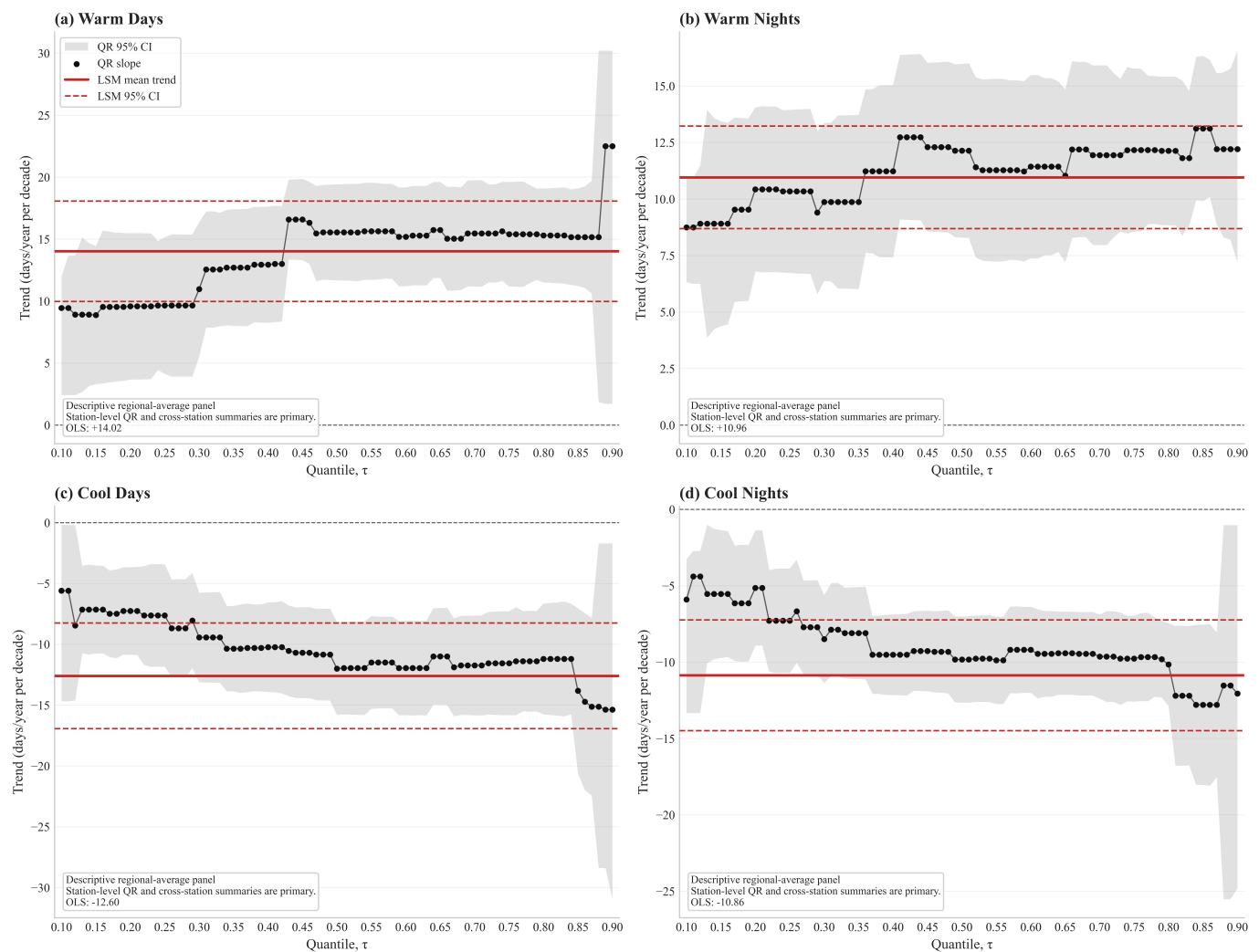
3.1. Overview

The results show a consistent directional pattern across the network, but they do not support a single uniform pattern of change. Warm indices generally rise, cool indices generally decline, and the magnitude of change varies systematically by quantile, index, and region. The most pronounced feature is the amplification of warm days at upper quantiles, whereas the most spatially persistent cooling-related feature is the contraction of cool days. Warm nights also increase across much of the country, but with flatter tail asymmetry and a more dominant single regional cluster. Cool nights decline widely as well, although their spatial organization is weaker than that of cool days.

At the station level, the directional consistency is pronounced. All 124 warm-day OLS slopes are positive, and 123/124 warm-day stations remain positive at both $q0.10$ and $q0.50$, with 122/124 still positive at $q0.90$. Warm nights are also broadly positive, though with more exceptions: 117/124 OLS slopes are positive, with 122/124, 117/124, and 116/124 positive at $q0.10$, $q0.50$, and $q0.90$, respectively. Cool days are almost uniformly negative, with all OLS, $q0.50$, and $q0.90$ slopes negative and only three stations showing positive $q0.10$ slopes. Cool nights are likewise predominantly negative, with 116/124 negative OLS slopes and negative Δt_1 at 116/124 stations. The central interpretation therefore rests on a broad distributional pattern rather than on a small number of unusual stations.

3.2. Regional synthesis of quantile-dependent change

Figure 3 and Table 1 bring the regional picture into focus by combining network-mean quantile profiles with station-level field-significance and dominant-cluster information. Four features stand out. Warm days show the strongest regional increase, with the slope rising from $+9.45$ days per decade at $q0.10$ to $+22.50$ at $q0.90$. Warm nights also increase across the distribution, but their upper-tail slope ($+12.21$) remains very close to their median slope ($+12.14$), which points to much weaker tail amplification than in warm days. Both cool indices decline regionally, with cool days showing the sharper contraction, especially at $q0.90$ (-15.37). The sign of $\Delta\tau_1$ is also revealing: it is positive for the warm indices and negative for the cool indices, indicating upper-tail amplification of warming-related frequencies but upper-tail contraction of cooling-related frequencies.



Panels summarize quantile-regression fits applied to annual station-mean series and are intended as descriptive pooled views only. Primary inference in the manuscript is based on station-level quantile regression, bootstrap summaries, and cross-station diagnostics.

Figure 3. Regional network-mean quantile-regression profiles for warm days, warm nights, cool days, and cool nights. Each panel is fitted to the annual network-mean series and is descriptive rather than a substitute for station-level inference.

Table 1. Regional synthesis of network-mean slopes, median-quantile field significance, and dominant-cluster behavior.

Index	OLS	q0.10	q0.50	q0.90	Delta1	FDR-retained stations at q0.50	Dominant cluster
Warm days	+14.02	+9.45	+15.55	+22.50	+13.06	115/124	Cluster 3 (n = 69, median Delta1 = +14.48)
Warm nights	+10.96	+8.74	+12.14	+12.21	+3.47	104/124	Cluster 4 (n = 95, median Delta1 = +7.74)
Cool days	-12.60	-5.60	-12.00	-15.37	-9.77	97/124	Cluster 2 (n = 74, median Delta1 = -10.47)
Cool nights	-10.86	-5.91	-9.83	-12.06	-6.16	86/124	Cluster 2 (n = 114, median Delta1 = -8.75)

The station-level synthesis closely mirrors the regional view. Delta1 is positive at 118/124 warm-day stations and 107/124 warm-night stations, but negative at all 124 cool-day stations and at 116/124 cool-night stations. Bootstrap tail inference adds further support to these directional changes: significant bootstrap intervals occur at q0.90 for 73 warm-day, 88 warm-night, 57 cool-day, and 61 cool-night stations, while q0.10 significance occurs at 47, 69, 74, and 68 stations, respectively. Field-significance control at q0.50 leads to the same broad conclusion, indicating that the regional signal is widespread rather than concentrated in only a handful of stations.

3.3. Day-night asymmetry

The day-night comparison adds an important layer to the physical interpretation. In this 1991-2024 network, the most pronounced warming signal is expressed during the day rather than at night. Warm nights do increase, but warm-day slopes are larger for every regional summary metric. The warm-night minus warm-day contrast is -3.06 for OLS, -0.70 at q0.10, -3.41 at q0.50, and -10.29 at q0.90. The corresponding Delta1 contrast is -9.59, indicating that upper-tail amplification is much more strongly expressed in warm days than in warm nights.

For the cooling-related pair, the same broad logic appears in reverse. Cool nights decline, but cool days decline more strongly, especially above the median. Cool-night minus cool-day contrasts are +1.74 for OLS, -0.30 at q0.10, +2.17 at q0.50, and +3.31 at q0.90, with a Delta1 contrast of +3.61. The strongest erosion of cool conditions is therefore concentrated in cool days, not cool nights, and the gap widens toward the upper part of the distribution.

Table 2. Regional day-night asymmetry, expressed as warm-night minus warm-day and cool-night minus cool-day.

Metric	Warm-night minus warm-day	Cool-night minus cool-day
OLS	-3.06	+1.74

Metric	Warm-night minus warm-day	Cool-night minus cool-day
q0.10	-0.70	-0.30
q0.50	-3.41	+2.17
q0.90	-10.29	+3.31
Delta1	-9.59	+3.61

This asymmetry matters because it suggests that daytime warm-extreme frequency is the clearest leading edge of recent thermal change in Iran, whereas nighttime warming, although widespread, is more evenly distributed across the quantiles. By the same token, the decline of cool conditions is more aggressive in the daytime indices, implying that the disappearance of unusually cool years is not merely a nocturnal phenomenon.

3.4. Representative stations and cluster-defined quantile geometries

Regionalization shows that the four indices do not organize in the same way. Warm days are distributed across four clusters with sizes 7, 47, 69, and 1, which points to substantial internal diversity. Warm nights are far more concentrated, with a dominant cluster of 95 stations. Cool days are mainly divided between two larger groups (45 and 74 stations), whereas cool nights are overwhelmingly concentrated in one cluster of 114 stations, with only a few highly atypical stations defining the remaining groups. This contrast in cluster balance is revealing: daytime and cool-day responses are more regionally differentiated, whereas cool-night and especially warm-night responses are more strongly shaped by a broad common pattern.

Table 3. Representative stations selected reproducibly as the observed stations closest to the baseline cluster centroids.

Index	Cluster	Representative station	Distance	q0.10	q0.50	q0.90	Delta1
Warm days	1	Garmsar	0.407	+4.44	+3.57	-0.00	-4.44
Warm days	2	Fasa	0.345	+7.14	+12.73	+13.94	+6.80
Warm days	3	Omidiyeh (Aghajari)	0.303	+10.00	+16.00	+27.14	+17.14
Warm days	4	Esfahan	0.000	+7.89	+13.33	+42.00	+34.11
Warm nights	1	Dorudzan	0.504	+1.18	-4.50	-4.50	-5.68
Warm nights	2	Abadan	0.423	+16.19	+16.67	+17.41	+1.22
Warm nights	3	Semnan	0.275	+5.36	+7.06	+7.00	+1.64
Warm nights	4	Bushehr (Coastal)	0.190	+8.18	+12.31	+15.20	+7.02
Cool days	1	Arak	0.333	-7.08	-12.11	-20.00	-12.92
Cool days	2	Gorgan	0.272	-5.00	-7.86	-15.00	-10.00
Cool days	3	Abumusa Island	0.495	-7.50	-20.00	-31.56	-24.06
Cool days	4	Chahbahar	0.834	-4.17	-9.50	-37.04	-32.87

Index	Cluster	Representative station	Distance	q0.10	q0.50	q0.90	Delta1
Cool nights	1	Dorudzan	0.138	+6.32	+9.23	+2.86	-3.46
Cool nights	2	Biyarjomand	0.102	-7.33	-9.64	-16.79	-9.45
Cool nights	3	Kish Island	0.000	-0.91	-14.44	-49.69	-48.78
Cool nights	4	Garmsar	0.000	-26.36	-19.57	-24.64	+1.72

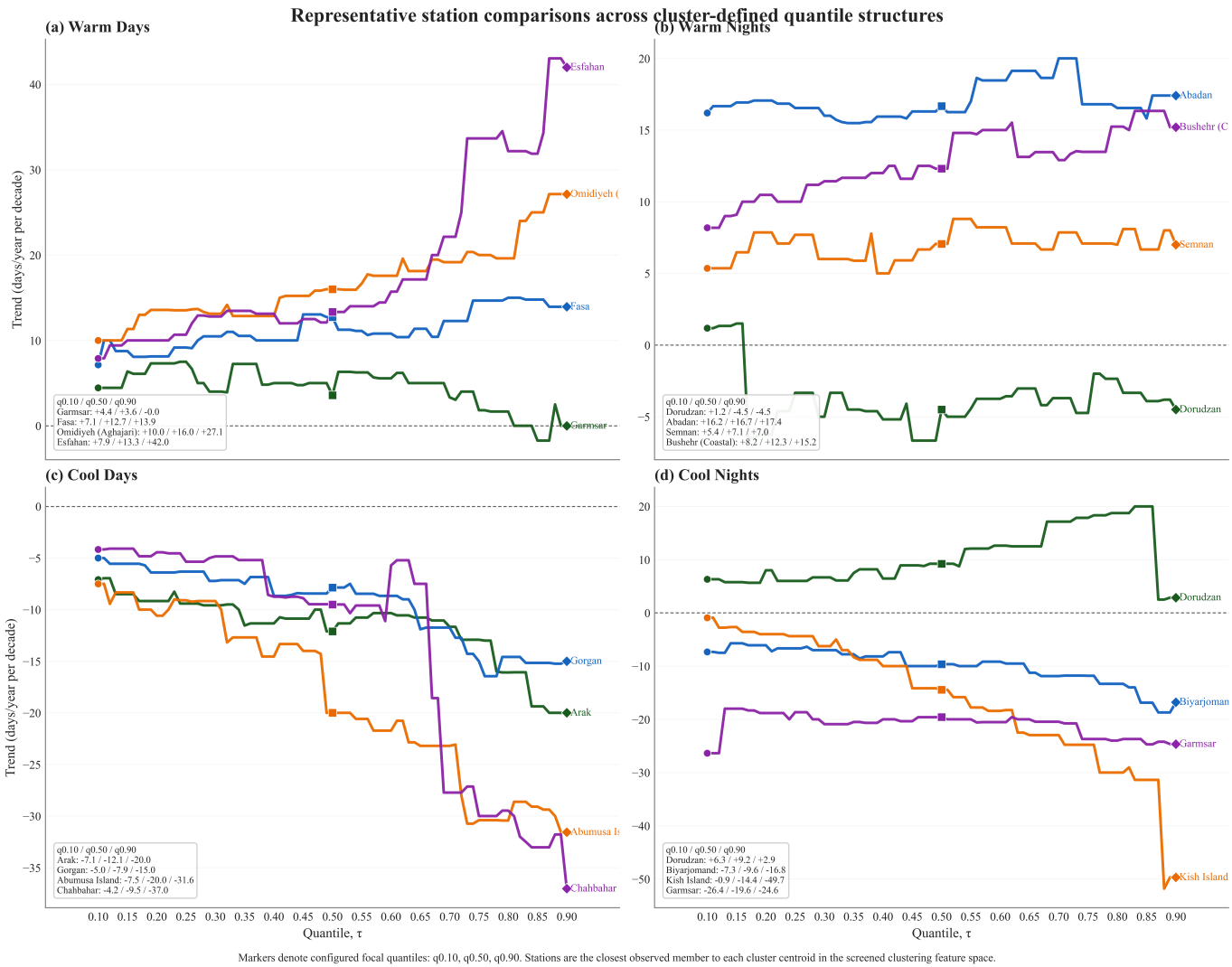


Figure 4. Representative-station panels for the four indices, with one station shown for each cluster according to the closest-observed-member rule.

The representative stations make the cluster geometry easier to interpret physically. For warm days, Esfahan defines a distinct single-station cluster with extreme upper-tail amplification ($\Delta 1 = +34.11$), while Garmsar captures the only representative warm-day geometry with a negative $\Delta 1$, indicating little or no upper-tail intensification. Warm-night behavior is more consolidated: Bushehr (Coastal) typifies the dominant positive pattern, whereas Dorudzan represents a minority cluster with negative median and upper-tail slopes. For cool days, Chahbahar and Abumusa Island illustrate the strongest upper-tail contraction, while Arak and Gorgan represent more moderate, but still negative, continental profiles. For cool nights, Biyarjomand captures the network's dominant declining pattern, Kish Island represents an extreme upper-tail-collapse outlier, and Garmsar stands out as a rare case with positive $\Delta 1$ because its lower-quantile decline is even stronger than its upper-quantile decline.

3.5. Spatial expression and regionalization

The spatial pattern of Δ_{11} indicates that distributional asymmetry is geographically organized rather than random. Warm-day Δ_{11} is positive at 118/124 stations, with the largest values at Esfahan (+34.11), Zanjan (+28.44), Daran (+28.15), Abumusa Island (+27.42), and Bandar-E-Anzali (+26.15). Warm-night Δ_{11} is also positive at most stations (107/124), but its geography is more mixed: large positive values occur at Zahak (+21.05), Khorramdareh (+17.23), Zanjan (+14.90), Nowshahr (+14.48), and Sarakhs (+14.23), while several stations show strongly negative values, including Ilam (-18.25) and Shiraz (-16.45). This more mixed pattern helps explain why warm nights rise strongly overall yet remain less sharply amplified in the upper tail than warm days.

For the cooling-related indices, the geographical pattern is even sharper. All 124 cool-day stations have negative Δ_{11} , with the most extreme contractions at Bandar-E-Lengeh (-33.70), Chahbahar (-32.87), Bushehr (Coastal) (-30.53), Orumiyeh (-26.39), and Jask (-25.54). Cool-night Δ_{11} is negative at 116/124 stations, and the most severe values occur at Kish Island (-48.78), Esfahan (-25.79), Sabzevar (-25.64), Shiraz (-24.75), and Abumusa Island (-24.27). These maps suggest that the most rapid erosion of cool extremes is concentrated in a subset of low-latitude, coastal, and inland-hot stations, rather than spread uniformly across the country.

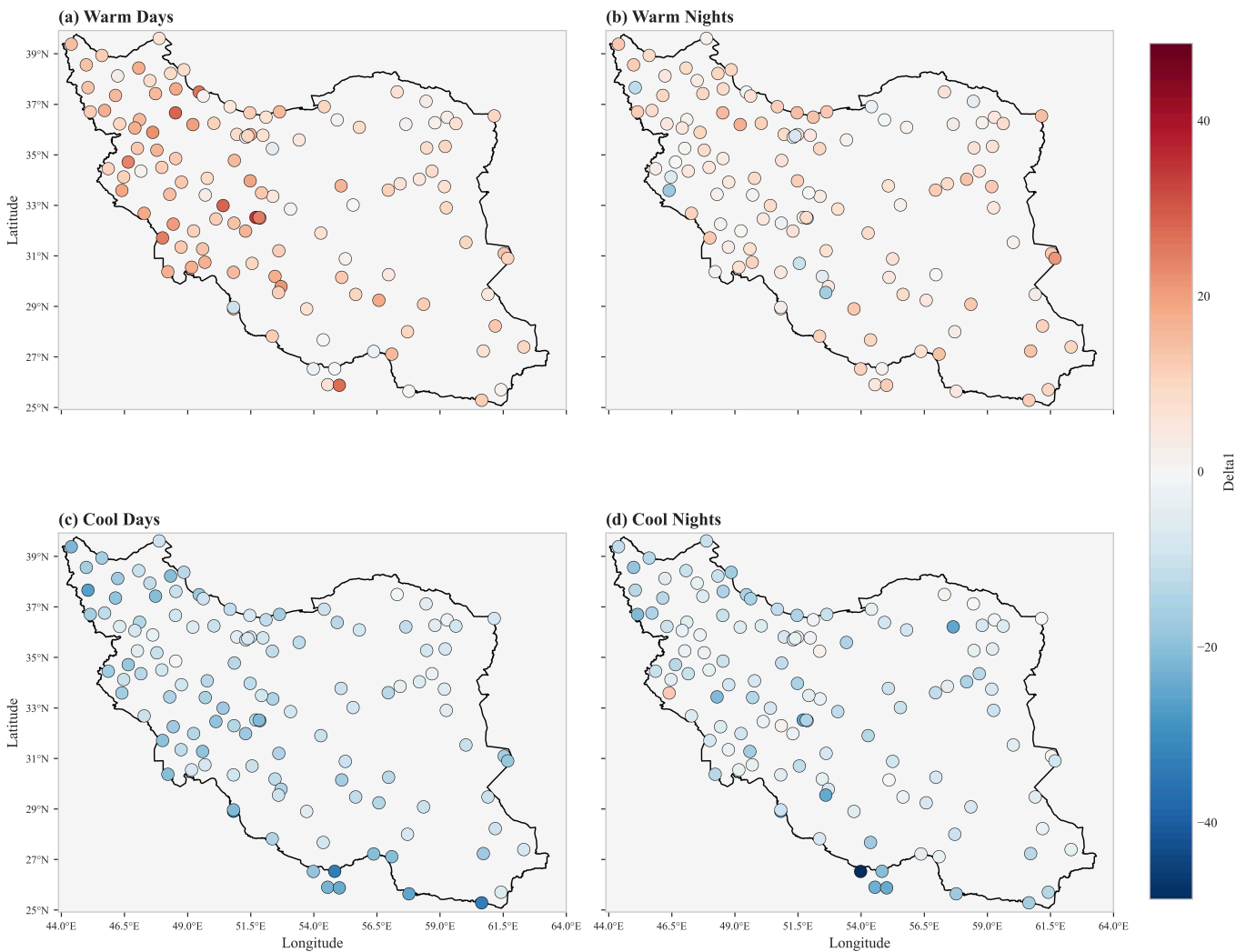


Figure 5. Station maps of $\Delta_{11} = \text{slope}(q_{0.90}) - \text{slope}(q_{0.10})$ for warm days, warm nights, cool days, and cool nights. Positive values indicate upper-tail amplification; negative values indicate relative upper-tail weakening or contraction.

Cluster size provides a second way of reading the same geography. The dominant warm-night cluster contains 95 stations and the dominant cool-night cluster contains 114 stations, implying that the night indices are shaped by large common regional modes with only a few atypical departures. Warm days and cool days, by contrast, are partitioned more evenly among multiple groups, which is consistent with stronger regional differentiation and with the more distinct day-side spatial patterns visible in Figure 5.

3.5.1. Additional structural and spatial diagnostics

The spatial diagnostics sharpen this interpretation. Moran's I is significant in 6/12 index-quantile fields. Cool days show the strongest and most consistent spatial organization, with Moran's I of 0.290 at $q0.10$, 0.192 at $q0.50$, and 0.348 at $q0.90$ (all $p = 0.002-0.004$). Warm days also show significant spatial structure at the tails (0.258 at $q0.10$, $p = 0.002$; 0.166 at $q0.90$, $p = 0.008$) but not at the median. Warm nights show significant lower-tail structure only (0.200, $p = 0.004$ at $q0.10$), whereas cool nights do not reach significance at $q0.10$ or $q0.50$ and are only marginal at $q0.90$ ($p = 0.058$). Spatial dependence is therefore evident, but it is specific to particular indices and quantiles rather than uniform across the full analysis.

The permutation-based spatial compactness test points in the same direction. Warm-day and cool-day clusters are more geographically compact than random labelings ($p = 0.002$ and 0.008 , respectively), whereas warm-night and cool-night clusters are not ($p = 0.882$ and 0.894). This suggests that day-index regionalization corresponds more closely to physically contiguous spatial groupings, while night-index clustering more often captures similar quantile geometry across geographically dispersed stations. Supplementary dendrograms and quantile-map panels reinforce the same point: station similarity changes across quantiles, and the apparent spatial field is more coherent for some indices and quantiles than for others. Supporting visual summaries are provided in Supplementary Figures S2-S7 and S12-S13, and tabular diagnostics are provided in Supplementary Tables S10-S11.

3.6. Geographical and elevational gradients

The driver analysis links these spatial contrasts to broad physiographic gradients. Warm-day $\Delta 1$ decreases significantly with longitude ($\text{std beta} = -0.395$, $p < 0.001$; $\rho = -0.426$), suggesting that the strongest upper-tail amplification tends to occur farther west in the network, although several southern stations also show large positive values. Warm-night $\Delta 1$ decreases with elevation ($\text{std beta} = -0.230$, $p = 0.012$; $\rho = -0.244$), which implies that upper-tail warm-night amplification is strongest at lower-elevation sites and weaker at highland stations.

Cool-day $\Delta 1$ shows the clearest geographical structure of all four indices. Its association with latitude ($\text{std beta} = +0.450$, $p < 0.001$), longitude ($+0.410$, $p < 0.001$), and elevation ($+0.263$, $p = 0.001$) implies that the most negative $\Delta 1$ values, and therefore the strongest upper-tail loss of cool days, occur preferentially at lower, more southerly, and more westerly stations. Cool-night $\Delta 1$ also becomes less negative with elevation ($\text{std beta} = +0.269$, $p = 0.003$; $\rho = +0.295$), again pointing to especially strong contraction of cool-night extremes in low-elevation settings.

These relationships point to a physically interpretable pattern. Low-elevation stations, many of them coastal or interior-hot, tend to show either stronger amplification of warm extremes or stronger contraction of cool extremes. Highland and northern stations also warm, but their distributional geometry is often flatter and less dominated by upper-tail extremes. These geographical controls do not explain all station-level behavior on their own, but they do suggest that the distributional trend structure is meaningfully conditioned by Iran's large-scale physiography rather than being arbitrary.

3.7. Robustness, methodological sensitivity, and temporal nonlinearity

The main directional conclusions remain stable under multiple sensitivity checks, although the magnitude of some summaries does depend on method. Excluding all 39 stations flagged by any detrended homogeneity diagnostic leaves 85 stations and changes the regional network-mean **Delta1** values only modestly: warm days shift from **+13.06** to **+13.05**, warm nights from **+3.47** to **+3.45**, cool days from **-9.77** to **-10.39**, and cool nights from **-6.16** to **-4.87**. This indicates that the sign structure and overall ordering are not driven by the flagged subset alone, even if the exact magnitude of cool-index asymmetry remains somewhat sensitive to station exclusion.

Bootstrap-depth sensitivity is similarly reassuring. Increasing the moving-block bootstrap from 200 to 400 replicates yields mean regional differences of only **-0.12** to **+0.09** across the four primary bootstrap mean metrics, with median absolute station-level differences mostly between **0.10** and **0.30** and station-to-station correlations above **0.996** in every case. By contrast, bootstrap-method sensitivity is more substantial in the tails: the mean absolute difference between moving-block and **meboot** summaries is about **0.97-1.16** days per decade at **q0.10** and **1.68-3.34** at **q0.90**. That does not undermine the sign of the results, but it does show that tail-uncertainty magnitudes depend materially on bootstrap design. Interpolation comparisons tell a similar story: mapped surfaces are more reproducible for some methods than for others. For example, linear RBF and linear interpolation agree strongly, with surface correlations of **0.84-0.93**, whereas thin-plate spline solutions diverge much more from those alternatives. This is another reason to interpret the maps as spatial displays rather than as independent inferential products.

Regionalization robustness is uneven, and that unevenness is itself informative. In the reduced-feature rerun, agreement with the baseline clustering is very high for cool nights (**ARI = 0.94**), moderate for warm nights (**0.49**), and low for warm days (**0.15**) and cool days (**0.06**). The alternative-method comparison, however, shows that warm-day regionalization remains reasonably reproducible across several clustering algorithms (**ARI = 0.63-0.98**); warm nights are highly stable under the average-cityblock alternative (**0.94**) but less so under complete linkage and k-means; cool days remain only moderately reproducible (**0.36-0.46**); and cool nights range from exact reproduction under the average-cityblock variant (**1.00**) to low agreement under other alternatives (**0.12-0.21**). The checks suggest that clustering is most stable for the dominant cool-night pattern and least uniquely determined for cool days, while warm-day clustering is sensitive to feature selection but still retains substantial structure across alternative hierarchical specifications.

No alternative reference-period comparison was generated by the original method-sensitivity module because its configured baseline and alternative settings collapse to the same effective period in the available dataset. The separate fixed-baseline climate-fingerprint analysis is reported below. Figure 6 therefore summarizes only the robustness components that produced distinct comparison outputs, namely bootstrap-method sensitivity, interpolation sensitivity, and clustering robustness.

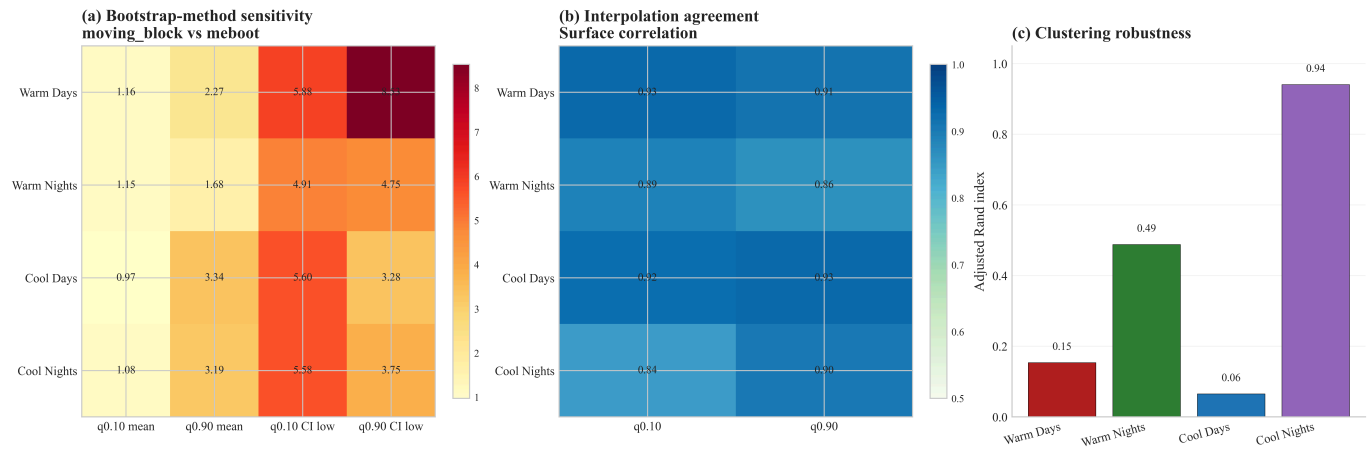


Figure 6. Multi-panel synthesis of the methodological-sensitivity and clustering-robustness analyses available for the present dataset.

The split-period analysis points to temporal nonlinearity rather than a simple story of stronger change in the later period at every quantile. For warm days, the regional OLS slope increases from **+10.94** in **1991–2007** to **+20.01** in **2008–2024**, and the median quantile also strengthens (**+10.43** to **+24.50**), but the upper-tail slope weakens (**+23.65** to **+14.97**). Warm nights show a similar contrast: the OLS slope rises from **+11.15** to **+14.22**, while the upper-tail slope falls from **+15.60** to **+6.36**. For the cool indices, the strongest negative OLS slopes occur in the earlier half of the record, decreasing from **–27.62** to **–10.91** for cool days and from **–21.83** to **–10.47** for cool nights. These results suggest that the 34-year record is not well described by a single monotonic acceleration narrative. Central and mean warming strengthen in the later half, but the tail geometry becomes more irregular, while the strongest cooling-related contractions are concentrated in the earlier subperiod.

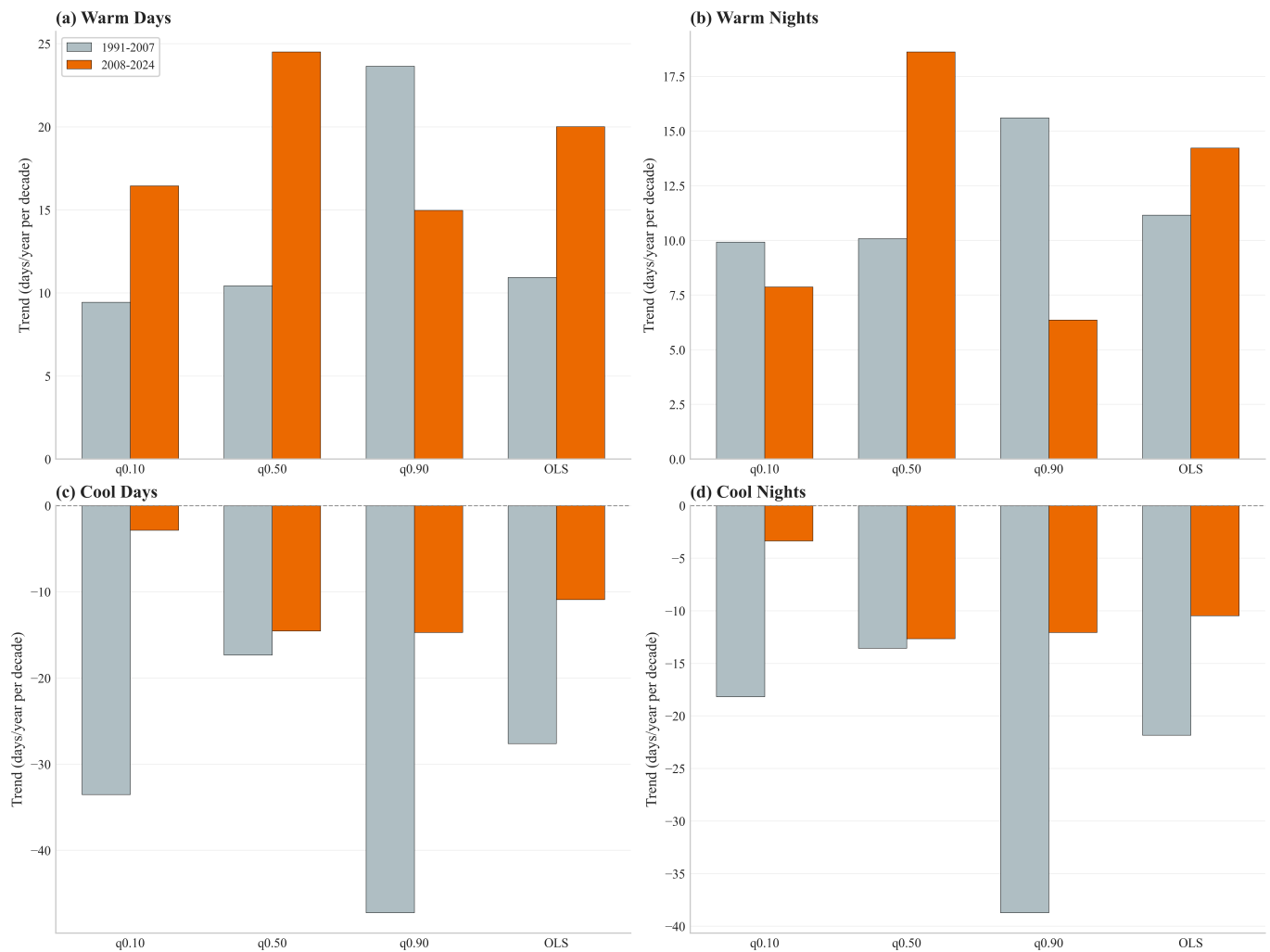


Figure 7. Comparison of regional network-mean $q0.10$, $q0.50$, $q0.90$, and OLS slopes between 1991-2007 and 2008-2024. The figure documents temporal nonlinearity rather than uniform amplification across every quantile.

The robustness analyses support the direction and rank ordering of the main conclusions while also clarifying their proper scope. The signs and relative ordering of the regional summaries are stable, but the exact magnitudes of tail contrasts and some clustering partitions should be interpreted as conditional on methodological choices rather than as uniquely fixed climatological constants. Supporting robustness outputs are compiled in Supplementary Figures S8-S11 and Supplementary Tables S4-S9.

3.8. Climate-change fingerprint and signal emergence

The QCF-EA results provide the main climate-change fingerprint test supporting the framework in the title. The station-derived regional temperature anomaly increased at $+0.446$ deg C per decade over 1991-2024 (95% CI: $+0.272$ to $+0.621$ deg C per decade), and the anomaly in 2024 was $+1.44$ deg C higher than in 1991. This regional warming covariate is not independent of the station archive, but it gives a physically interpretable scale against which to express the extreme-index changes. The correct reading is therefore one of internally consistent observational co-evolution: the anomaly and the thermal extremes move together, but the analysis is not designed to identify an external forcing mechanism.

The fixed-baseline analysis gives the clearest frequency-based evidence of warming-related restructuring. Using 1991-2007 thresholds for all years, the 2008-2024 period contains, on average, $+36.78$ warm days and $+28.41$ warm nights per year relative to the baseline period. Cool extremes move in the opposite direction,

with **-10.69** cool days and **-8.19** cool nights per year. The station-level consistency is also strong: the shift is warming-consistent at **124/124** stations for warm days, **117/124** for warm nights, **124/124** for cool days, and **110/124** for cool nights. These values show that the main results are not an artifact of defining thresholds over the full 1991-2024 period.

The warming-linked quantile-response models point to the same interpretation. At the network level, the **q0.90** response to one degree of regional station warming is **+29.62** warm days, **+22.60** warm nights, **-21.03** cool days, and **-19.66** cool nights per year. Median responses are also directionally coherent: **+40.41**, **+25.42**, **-16.51**, and **-14.51** days per year per degree Celsius for warm days, warm nights, cool days, and cool nights, respectively. At the station level, the **q0.50** warming response has the expected sign at all 124 warm-day, warm-night, and cool-day stations, and at **119/124** cool-night stations.

Table 4. Climate-fingerprint diagnostics linking extreme-index changes to fixed-baseline shifts, regional warming, and signal emergence.

Index	Fixed-baseline shift, 2008-2024 minus 1991-2007	Direction-consistent stations	Network q0.90 response to warming	q0.50 emerged stations
Warm days	+36.78 days/year	124/124	+29.62 days/year per deg C	99/124
Warm nights	+28.41 days/year	117/124	+22.60 days/year per deg C	97/124
Cool days	-10.69 days/year	124/124	-21.03 days/year per deg C	69/124
Cool nights	-8.19 days/year	110/124	-19.66 days/year per deg C	68/124

Signal emergence is strongest for the median warm indices. At **q0.50**, the bootstrap signal-to-noise threshold is exceeded at **99/124** warm-day stations and **97/124** warm-night stations, compared with **69/124** cool-day and **68/124** cool-night stations. Emergence counts are lower for **Delta1** because tail-contrast uncertainty is wider, but the direction of **Delta1** remains strongly consistent with the warming fingerprint. This distinction is important: the evidence for directional warming-related change is stronger than the evidence that every station has a precisely estimated asymmetry magnitude.

The integrated climate-fingerprint score is **0.897**, summarizing directionality, fixed-baseline shifts, warming-linked responses, signal emergence, FDR field significance, homogeneity-exclusion stability, and bootstrap-depth stability. The network quantile-fingerprint score is **1.000**: all four indices show the expected signs across **q0.10**, **q0.50**, **q0.90**, and **Delta1**. Under 499 circular-shift permutations, this coherence is unusual under the temporal-shift null ($p = 0.002$). Figure 8 summarizes the component scores and the permutation test. These findings are best read as statistically consistent signal detection within an observational archive, not as formal anthropogenic attribution, although they do support a warming-consistent interpretation of the recent Iranian station record.

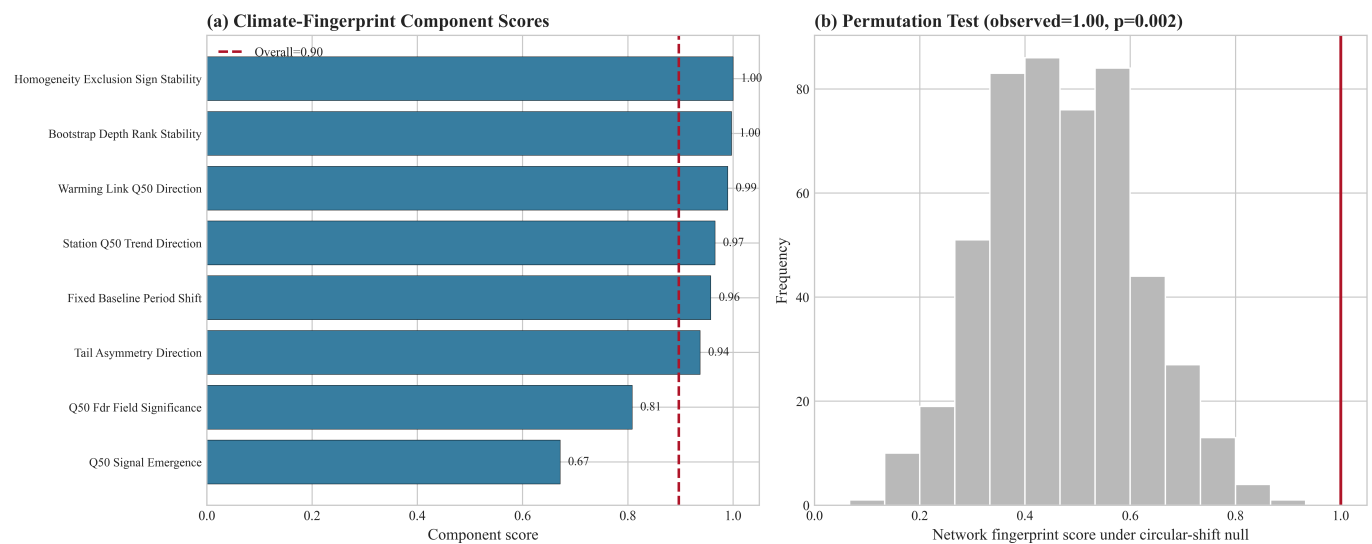


Figure 8. Integrated climate-fingerprint component scores and circular-shift permutation test for the network quantile fingerprint. The component score summarizes consistency across trend direction, tail asymmetry, fixed-baseline shifts, warming-linked responses, signal emergence, field significance, and robustness diagnostics.

3.9. Climate-regime stratification of quantile change

The Köppen-Geiger stratification shows that the warming-related fingerprint is not confined to a single climatic setting. The 124 stations fall into six analysis regimes: 36 hot-desert (BWh), 18 cold-desert (BWk), 10 hot-steppe (BSh), 35 cold-steppe (BSk), 15 temperate (Csa/Cfa), and 10 cold dry-summer (Dsa) stations. Mean classification confidence is generally high (76.2–91.1% across regimes), although confidence is lower in the colder arid regimes where Köppen-Geiger boundaries are more transitional. This structure gives a useful climate-context layer: the station network spans low-elevation hot deserts and hot steppes, higher cold deserts and cold steppes, temperate northern/western settings, and the highest-elevation cold dry-summer group.

The sign of change is broadly coherent across all regimes, but the quantile geometry differs. Warm-day upper-quantile amplification is strongest in hot-steppe stations, where the mean $q0.90$ slope reaches +23.85 days per decade, followed by cold dry-summer (+22.23) and cold-steppe (+21.62) stations. The strongest cool-day upper-quantile contraction occurs in hot-desert stations (−20.59 days per decade), with cold dry-summer (−19.62) and temperate (−18.37) stations also showing strong contraction. Warm-night upper-quantile change is largest in hot deserts (+16.64 days per decade), whereas cool-night upper-quantile contraction is also strongest in hot deserts (−18.36). Thus, arid lowland regimes show especially strong night and cool-day erosion, while steppe and cold dry-summer regimes show particularly strong warm-day amplification.

Table 5. Köppen-Geiger climate-regime summaries for selected quantile and fixed-baseline diagnostics.

Climate regime	Stations	Mean elevation	Warm-day $q0.90$ slope	Cool-day $q0.90$ slope	Fixed warm-day shift	Fixed cool-day shift	Fingerprint score
BWh hot desert	36	575 m	+15.84	−20.59	+32.01	−11.12	0.900

Climate regime	Stations	Mean elevation	Warm-day q0.90 slope	Cool-day q0.90 slope	Fixed warm-day shift	Fixed cool-day shift	Fingerprint score
BWk cold desert	18	1402 m	+16.75	-15.28	+28.73	-8.04	0.847
BSh hot steppe	10	495 m	+23.85	-16.85	+46.62	-11.98	0.892
BSk cold steppe	35	1442 m	+21.62	-17.56	+38.36	-10.59	0.887
C temperate	15	672 m	+20.44	-18.37	+41.12	-10.13	0.889
Dsa cold dry-summer	10	1917 m	+22.23	-19.62	+46.63	-13.81	0.925

The fixed-baseline and warming-response diagnostics reinforce this climate-regime interpretation. Relative to 1991–2007 thresholds, the largest warm-day shifts occur in cold dry-summer and hot-steppe stations (+46.63 and +46.62 days/year), while the smallest occur in cold deserts (+28.73). Cool-day losses are strongest in cold dry-summer stations (-13.81 days/year) and weakest in cold deserts (-8.04). Warming-linked responses show the same contrast: the q0.50 warm-day response to one degree of regional station warming reaches +48.39 days/year per deg C in cold dry-summer stations, compared with +29.62 in cold deserts, and the q0.90 cool-day response is most negative in cold dry-summer stations (-29.74 days/year per deg C). After FDR correction, the clearest between-regime contrasts are the warming-linked q0.50 warm-day response and the warming-linked q0.90 cool-day response ($q = 0.048$ for both); several raw contrasts in warm-day trends and fixed-baseline shifts are suggestive but do not remain below the 0.05 FDR threshold.

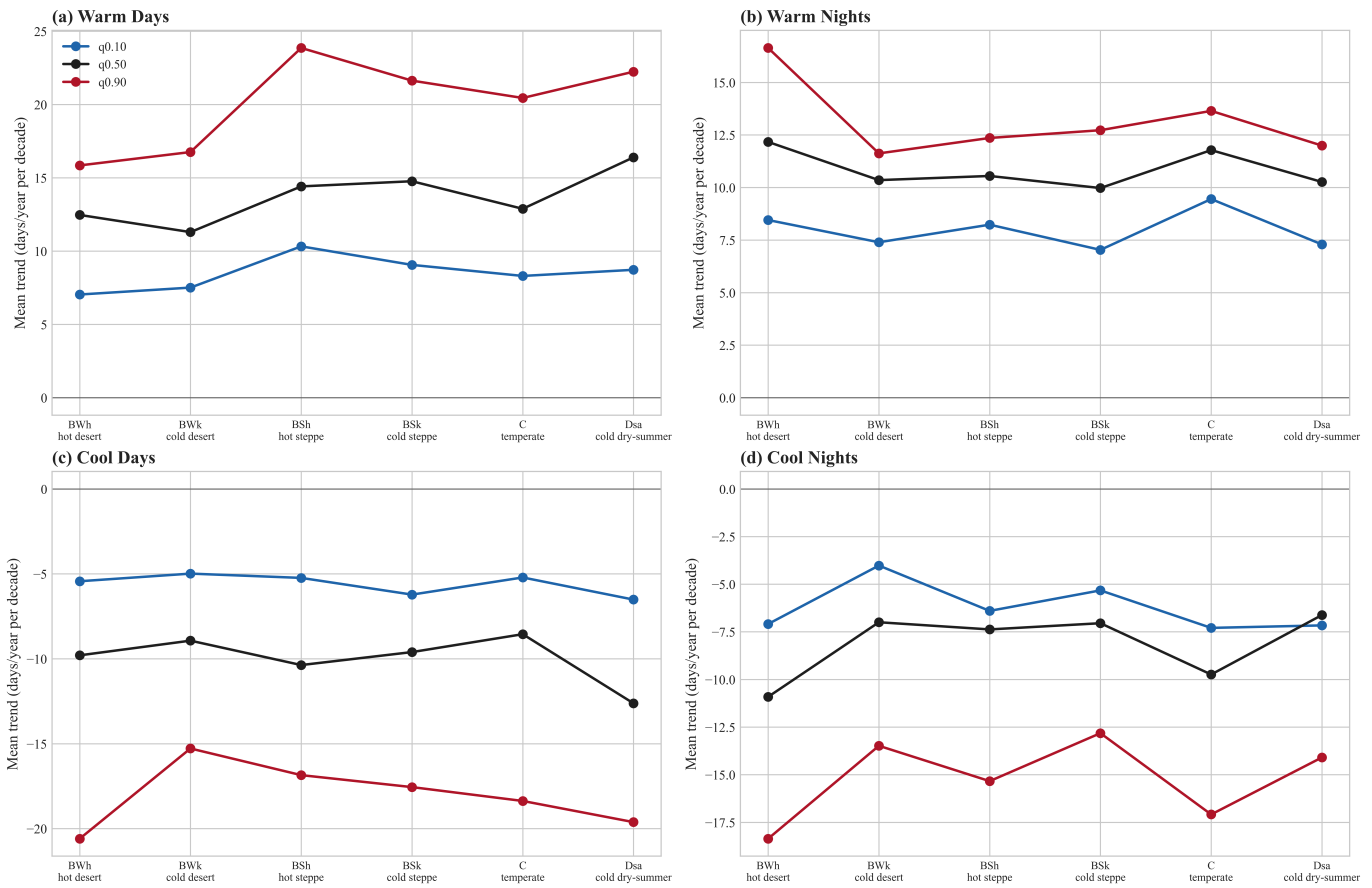


Figure 9. Köppen-Geiger climate-regime mean quantile slopes for warm days, warm nights, cool days, and cool nights. Values summarize station-level slopes within each regime; they are intended to compare climate-regime structure rather than replace station-level inference.

The regime-level fingerprint scores remain high in every climate setting, ranging from **0.847** in cold-desert stations to **0.925** in cold dry-summer stations. This matters for interpretation. The climate-regime analysis does not weaken the national warming pattern; instead, it shows that the warming fingerprint takes different quantile forms in different climatic environments. This reading is consistent with recent evidence that Iran's Köppen-Geiger climate types are themselves undergoing warming-driven transitions toward warmer and drier regimes (Jamali et al., 2026). Hot deserts carry the strongest warm-night and cool-day upper-tail patterns, hot steppes and cold dry-summer stations carry the strongest warm-day amplification, and cold deserts show the weakest but still directionally consistent fingerprint. The result is a two-level conclusion: recent thermal-extreme change in Iran is nationally consistent in sign, but climatically differentiated in magnitude and distributional shape. The next step is therefore not a separate topic but an impact-relevant extension of the same signal: if upper-tail warm-day amplification is a leading feature of the thermal record, then a key hydroclimatic question is whether those increasingly hot years are also becoming dry across a larger share of the station network.

3.10. Compound dry-hot concurrence and spatial connectedness

The empirical-copula layer shows that the temperature signal is accompanied by a marked increase in the spatial extent of compound dry-hot station-years. Under the annual definition, **RP** $\geq$ **10** dry-hot events affected an average of **12.71** stations in **1991-2007** and **39.82** stations in **2008-2024**, a late-minus-early increase of **27.12** stations. The trend is positive under both Kendall's rank test ($\tau = 0.318$, $p = 0.0086$) and a linear count slope (**+15.63** stations per decade, $p = 0.0031$). The rarer annual **RP** $\geq$ **20** events show the same direction, rising from **3.12** to **17.12** affected stations ($\tau = 0.320$, $p = 0.0106$).

The warm-season definition gives an even stronger signal. $RP \geq 10$ dry-hot events based on June–September precipitation and maximum temperature affected 10.35 stations on average in 1991–2007, but 41.53 stations in 2008–2024. The corresponding Kendall trend is $\tau = 0.583$ ($p < 0.001$), and the linear slope is +19.24 stations per decade. For $RP \geq 20$, the affected-station count increased from 2.59 to 16.12 stations ($\tau = 0.539$, $p < 0.001$; linear slope +8.72 stations per decade). The most extensive warm-season $RP \geq 10$ year is 2021, when 99 valid stations crossed the empirical joint-return-period threshold. Several other recent years, including 2019, 2023, 2024, 2017, and 2013, also rank among the most spatially extensive dry-hot years in the station archive.

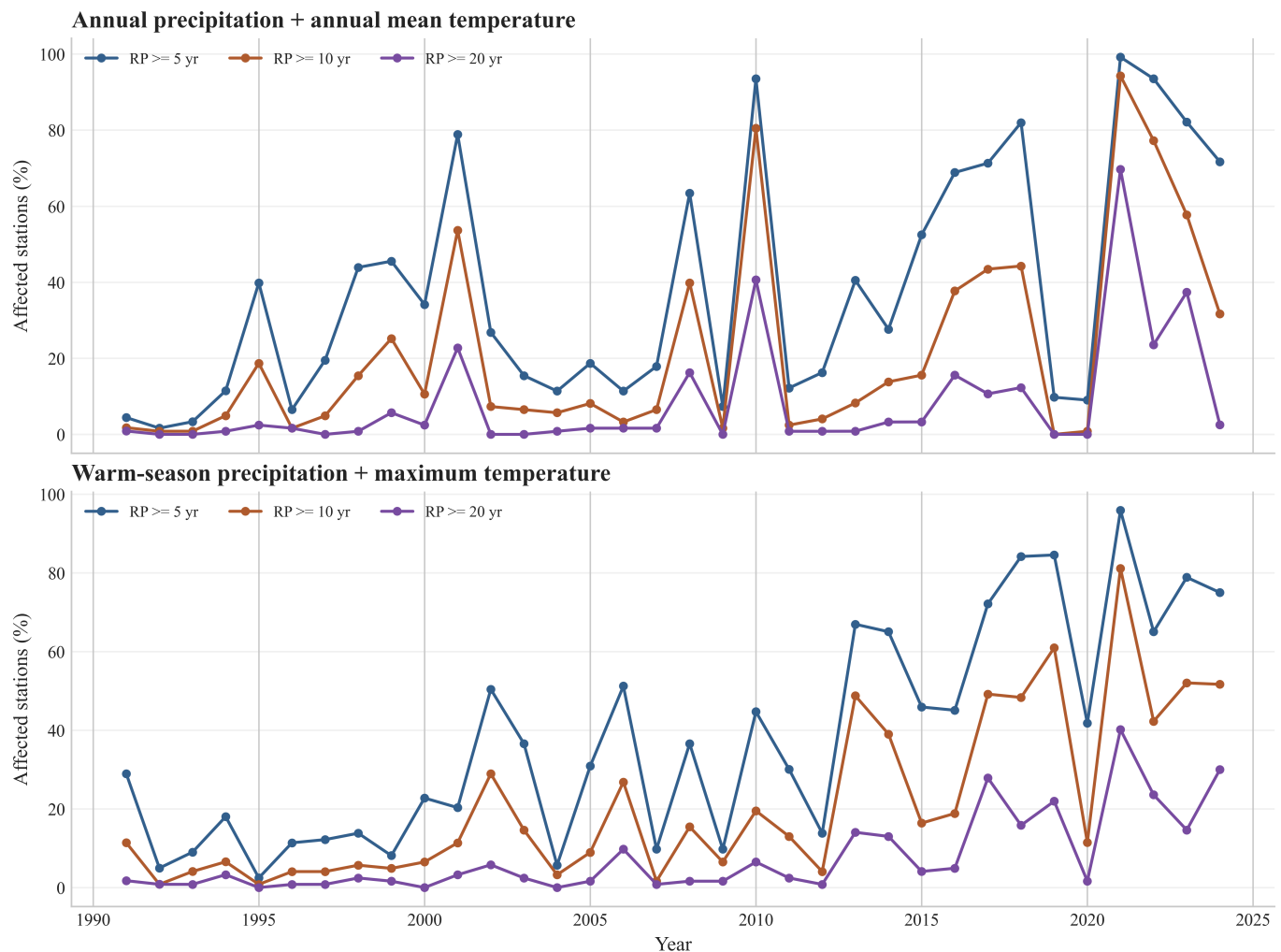


Figure 10. Annual percentage of valid stations experiencing compound dry-hot events under the empirical *AND* return-period definition. The annual definition pairs total annual precipitation with annual mean temperature; the warm-season definition pairs June–September precipitation with June–September maximum temperature.

Table 6. Compound dry-hot affected-station trends and early-late contrasts.

Definition	Return-period threshold	Mean affected stations, 1991-2007	Mean affected stations, 2008-2024	Kendall tau	Kendall p	Linear slope
Annual precipitation + annual mean temperature	RP $\geq$ 10	12.71	39.82	0.318	0.0086	+15.63 stations/decade
Annual precipitation + annual mean temperature	RP $\geq$ 20	3.12	17.12	0.320	0.0106	+7.95 stations/decade
Warm-season precipitation + maximum temperature	RP $\geq$ 10	10.35	41.53	0.583	<0.001	+19.24 stations/decade
Warm-season precipitation + maximum temperature	RP $\geq$ 20	2.59	16.12	0.539	<0.001	+8.72 stations/decade

Early-late distribution tests support the same interpretation. For the warm-season RP $\geq$ 10 affected-station percentage, the mean network extent increased from 8.50% to 34.04% of valid stations, and the two periods differ under Kolmogorov-Smirnov ($p = 0.00024$), Cramer-von Mises ($p = 0.000057$), and Anderson-Darling ($p < 0.001$) tests. The warm-season RP $\geq$ 20 distribution also shifts upward (2.12% to 13.22%; KS $p = 0.0046$; Cramer-von Mises $p = 0.00057$; Anderson-Darling $p < 0.001$). Annual dry-hot extent changes are less abrupt than warm-season changes but remain statistically visible at both RP $\geq$ 10 and RP $\geq$ 20.

The station-frequency maps show that the compound signal is not confined to one isolated part of the network. Warm-season RP $\geq$ 10 event frequency in 2008-2024 exceeds 50% at several stations, including Ravansar, Yasuj, Dogonbadan, Torbat-E Heydariyeh, Khash, Tehran (Geophysic), Tehran (Shemiran), and Kerman. The largest late-minus-early increases are likewise concentrated at stations where the early-period frequency was low or moderate but the late-period frequency rose sharply. These maps should be read as station-based frequency displays, not as gridded area estimates.

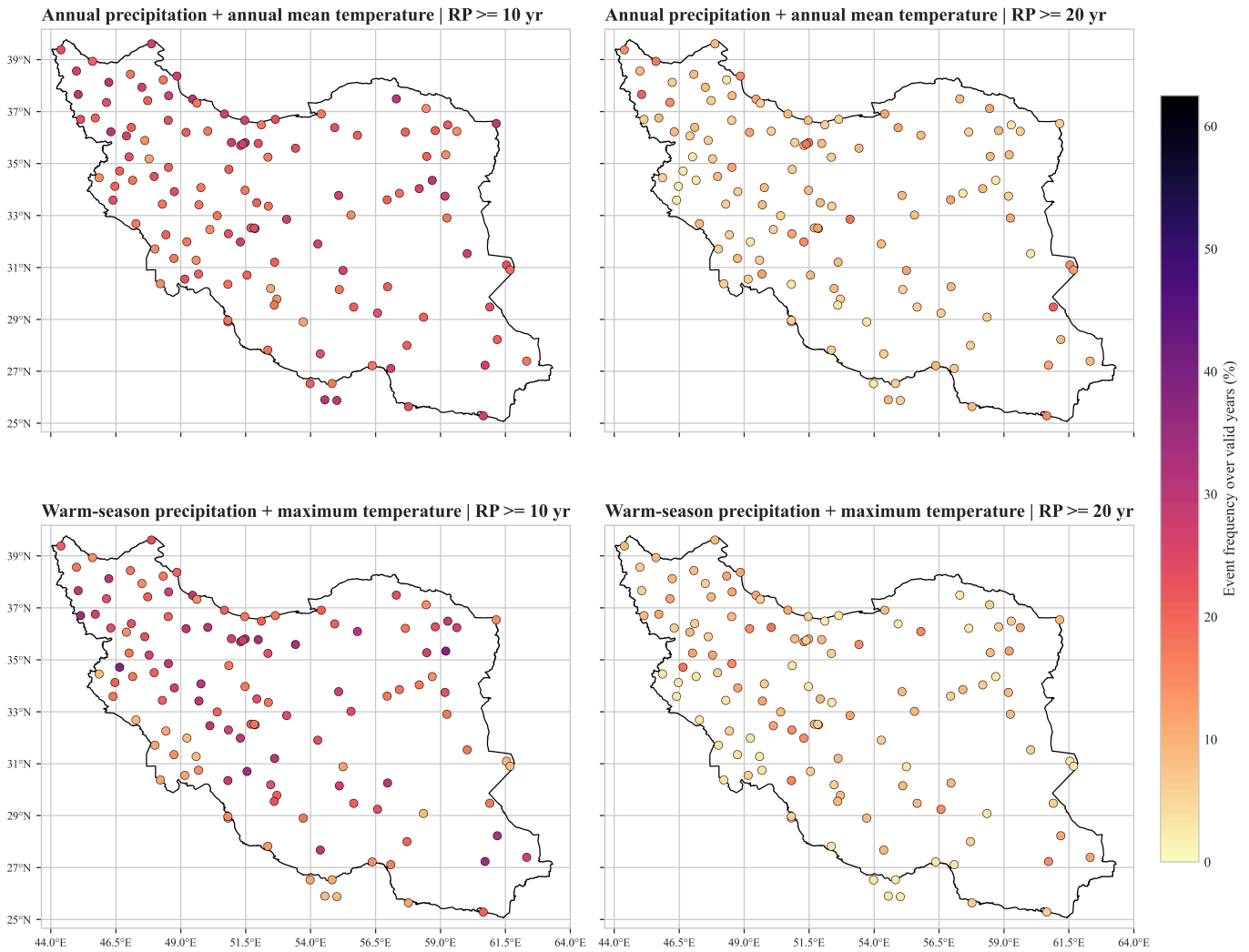


Figure 11. Station-level frequency of compound dry-hot events for the annual and warm-season definitions at $RP \geq 10$ and $RP \geq 20$. Values are percentages of valid station-years.

The dominant-driver summary adds a physically useful interpretation. In the warm-season $RP \geq 10$ event set, early-period station-events are mostly dry-dominant (147 dry-dominant, 27 hot-dominant, and 2 co-dominant events). In 2008–2024, the balance reverses: 535 station-events are hot-dominant, compared with 141 dry-dominant and 30 co-dominant events. The rarer warm-season $RP \geq 20$ events show an even sharper version of this shift, from 39 dry-dominant and 4 hot-dominant events in 1991–2007 to 51 dry-dominant and 205 hot-dominant events in 2008–2024. This pattern is consistent with the broader warming signal in the thermal indices: recent compound dry-hot years are increasingly defined by heat excess rather than by precipitation deficit alone.

Spatial connectedness also strengthens in the warm-season compound fields. The annual Moran's I series for warm-season $RP \geq 10$ events has a positive slope of +0.075 per decade ($p = 0.0047$; Kendall $p = 0.0095$), while $RP \geq 20$ events show a slope of +0.087 per decade ($p = 0.0017$; Kendall $p = 0.0016$). These values suggest that rare warm-season dry-hot events are not only becoming more frequent across the station network; they are also becoming more spatially clustered. Annual dry-hot connectedness is weaker and less statistically stable, which reinforces the interpretation that the compound signal is most sharply expressed during the hot season.

Taken together, the compound dry-hot layer strengthens the manuscript in two ways. First, it shows that the warming-related thermal fingerprint has a hydroclimatic counterpart: heat and precipitation deficit are increasingly co-occurring across the Iranian station network. Second, it clarifies the impact-relevant context of

the quantile results. The upper-tail amplification of warm days is not merely a statistical feature of annual temperature-index distributions; in recent years it is increasingly embedded in broader dry-hot station-year conditions, especially during the warm season.

3.11. Comparison with the wider climatological literature

The results align with the broader literature in several ways, while also sharpening the picture for Iran. The widespread increase in warm-event frequencies and decline in cool-event frequencies are consistent with prior evidence of intensifying heat extremes across Iran and the wider MENA domain (Zhang et al., 2005a; Soltani et al., 2016; Vaghefi et al., 2019; Zittis et al., 2016; Francis and Fonseca, 2024). They also complement recent Iran-focused projection studies that indicate increasing hydro-climatic and climate-extreme risks under warming (Naderi et al., 2024; Jahanshahi et al., 2024). Likewise, the strong contraction of cool days and cool nights is consistent with the global reduction of cold extremes documented in large observational syntheses (Alexander et al., 2006; Donat et al., 2013; Dunn et al., 2020). The dry-hot results connect this temperature-focused evidence to the compound-event literature, which emphasizes that the dependence between drivers can reshape risk more strongly than either driver in isolation (Leonard et al., 2014; Zscheischler and Seneviratne, 2017; Zscheischler et al., 2018). The observed shift toward hot-dominant compound station-events is also consistent with studies showing that warming can become an increasingly important component of dry-hot extremes even when precipitation changes alone are less spatially uniform (Sarhadi et al., 2018; Alizadeh et al., 2020).

What stands out more distinctly in the present analysis is the degree of distributional asymmetry and the fact that, in this dataset, recent warming is more strongly expressed in daytime warm extremes than in nighttime ones. That pattern is physically plausible in an arid and topographically complex setting where land-surface feedbacks, cloud variability, and radiative conditions can favor strong daytime amplification (Seneviratne et al., 2010; Sanchez-Lorenzo et al., 2017; Perkins-Kirkpatrick and Lewis, 2020). More recent studies reinforce this interpretation by showing that hot extremes can become drier in arid regions, that soil heat extremes may outpace near-surface air-temperature extremes, and that circulation persistence, circulation-model mismatch, and event-scale soil moisture-temperature coupling can all modulate heat extremes beyond what would be expected from mean warming alone (McKinnon et al., 2021; Rousi et al., 2022; Vautard et al., 2023; Maraun et al., 2025; García-García et al., 2023; Lv et al., 2026). At the same time, the concentration of strong warm-night amplification and strong cool-night contraction at low elevations is consistent with the growing importance of nocturnal and humid-heat risk in southern and coastal southwest Asia (Pal and Eltahir, 2016; Raymond et al., 2020; Raymond et al., 2024; Perkins-Kirkpatrick et al., 2026).

The results also fit the broader quantile-regression literature. Earlier studies showed that quantile-specific temperature trends can reveal structures that OLS or mean-only analysis tends to hide (Barbosa et al., 2011; Fan, 2014). The present study extends that logic to a dense national station network in Iran and shows that the same principle matters strongly in a dryland setting: warm-day amplification, warm-night flattening, and uneven cool-extreme contraction would all be only partly visible in mean trends alone. The QCF-EA layer adds a further methodological contribution by translating the distributional trend structure into a reproducible observational fingerprint, while the Köppen-Geiger stratification links that fingerprint to internationally recognizable climate regimes (Beck et al., 2018). In that sense, the study adds regional specificity to the wider literature by showing that recent thermal-extreme change across Iran is not just warming in sign, but also differentiated in shape and supported by multiple signal-detection checks.

3.12. Limitations

Several limitations define the scope of inference. The record length is 34 years, which is adequate for a consistent recent-trend analysis but shorter than ideal for very stable tail estimation or for strong claims about multidecadal nonlinearity. The main thermal indices are built with thresholds derived from the available 1991–2024 record rather than from an independent earlier climatological normal, so those annual counts are distribution-relative within the analyzed period. The fixed-baseline QCF-EA check partly addresses this concern by using 1991–2007 thresholds, but that baseline is itself short and should not be treated as a full pre-change climatology. The same caution applies to the empirical dry-hot return periods. With 34 years of data, a station-level nonparametric return period cannot support stable claims about 50- or 75-year compound events; this is why the analysis centers on $RP \geq 10$ and $RP \geq 20$ thresholds and treats them as empirical rarity classes within the observed archive. In addition, the daily records were screened for quality and homogeneity but were not formally homogenized with a metadata-supported daily adjustment procedure, which means that some local tail magnitudes may still reflect unresolved non-climatic shifts. For that reason, the climate-fingerprint and dry-hot outputs should be understood as robust descriptive concordance measures, not as substitutes for independent attribution evidence.

The main trend framework is station-wise, not fully spatial, so covariance enters the analysis through diagnostics and regionalization rather than through a hierarchical spatial quantile model (Reich, 2012). The analytic tail outputs were also insufficient for national FDR screening at $q0.10$ and $q0.90$, which means that field-significance interpretation is anchored mainly in the median quantile while the tails rely on bootstrap intervals. Several clustering results are exploratory by design, and their stability varies by index; the night-index clusters, in particular, are not especially geographically compact relative to random labelings. The driver analysis is intentionally simple and does not directly model circulation, soil moisture, humidity, radiation, or land-use effects. The dry-hot module uses precipitation totals and temperature means/maxima; it does not include soil moisture, vapor-pressure deficit, evaporative demand, irrigation, groundwater use, or crop exposure, so it should not be read as a complete drought-impacts analysis. Its spatial extent metrics are also station-count and station-frequency measures rather than true affected-area estimates. The Köppen-Geiger stratification is useful for climate-context interpretation, but it is based on an external 1980–2016 climatological map and cannot resolve all station microclimates, local siting effects, urbanization, or sub-grid coastal and mountain gradients. Some regimes also have modest sample sizes ($n = 10\text{--}18$), and the Cfa stations had to be merged with Csa stations to avoid unstable two-station summaries. The regional warming covariate used in QCF-EA is derived from the same station archive as the extreme indices, so it is best interpreted as a physically scaled observational association rather than as an independent attribution forcing. Finally, interpolated surfaces are used for display rather than as primary evidence, the split-period comparison is descriptive rather than a formal breakpoint-detection framework, and the circular-shift fingerprint test evaluates temporal coherence rather than human-caused attribution.

None of these limitations overturn the main conclusions, but they do help define their proper scope. The study is best read as a station-based synthesis of recent distributional change, climate-regime differentiation, warming-related observational signal emergence, and compound dry-hot concurrence across Iran. It is not a complete attribution analysis, a drought-impact assessment, or a fully spatially explicit quantile model. The most useful next steps would combine metadata-supported daily homogenization, explicit spatial quantile modeling, physically richer diagnostics of circulation, radiation, humidity, evaporative demand, soil moisture, and land-surface processes, and comparison with reanalysis or climate-model counterfactuals. This extension is especially important because recent work shows that circulation changes and precipitation-soil-moisture coupling can materially shape the amplitude and predictability of heat extremes (Vautard et al., 2023; Maraun et al., 2025; Lv et al., 2026).

4. Conclusions

This study shows that recent thermal-extreme change across Iran is fundamentally distributional, not merely mean-based. Using 124 meteorological stations for 1991-2024, the analysis indicates that warm days provide the strongest warming-related pattern. Regional network-mean slopes increase from $+9.45$ days per decade at $q0.10$ to $+22.50$ at $q0.90$, yielding $\Delta1 = +13.06$. Warm nights also increase across the country, but they do so more evenly across the distribution ($+8.74$, $+12.14$, and $+12.21$ days per decade at $q0.10$, $q0.50$, and $q0.90$; $\Delta1 = +3.47$). Cool days and cool nights both decline. The upper-quantile contraction is stronger for cool days (-5.60 , -12.00 , -15.37 ; $\Delta1 = -9.77$) than for cool nights (-5.91 , -9.83 , -12.06 ; $\Delta1 = -6.16$). The central message is therefore not simply that Iran has warmed. It is that the internal shape of annual thermal-extreme distributions has changed in a systematic and index-specific way.

Several aspects of the study deserve emphasis. It provides a dense station-based national assessment in which inference remains anchored in observations rather than interpolated fields. Its main novelty is the construction of a single reproducible climate-change fingerprint framework that combines quantile regression, bootstrap uncertainty estimation, field-significance screening, spatial autocorrelation diagnostics, cluster-based regionalization, climate-regime stratification, and compound dry-hot concurrence. It also shows that the main results are not confined to a small subset of stations: at $q0.50$, FDR control retained $115/124$ warm-day, $104/124$ warm-night, $97/124$ cool-day, and $86/124$ cool-night station results. Finally, it reveals that spatial organization differs by index. Cool days show the most persistent spatial autocorrelation, whereas warm days exhibit the clearest upper-tail amplification and the strongest evidence of geographically compact cluster structure. Warm nights and cool nights, by contrast, are characterized by larger dominant clusters, indicating broad common modes with fewer highly atypical departures.

The climate-fingerprint layer supports this interpretation. The station-derived regional temperature anomaly increased by $+0.446$ deg C per decade. Fixed-baseline indices show large late-period shifts relative to 1991-2007: $+36.78$ warm days, $+28.41$ warm nights, -10.69 cool days, and -8.19 cool nights per year. The network $q0.90$ response to regional warming has the expected direction across all four indices, and the integrated fingerprint score reaches 0.897 . The network quantile-fingerprint score is 1.000 under the circular-shift null ($p = 0.002$). These results support a cautious observational inference: recent thermal extremes in Iran carry a warming-consistent pattern. They do not, by themselves, constitute formal anthropogenic attribution.

The Köppen-Geiger stratification adds a second conclusion: the fingerprint appears across climate regimes, but its form differs by regime. Hot-steppe and cold dry-summer stations show the strongest warm-day amplification, hot deserts show the strongest cool-day and warm-night upper-quantile patterns, and all six regimes retain high fingerprint scores ($0.847-0.925$). Iran's recent thermal-extreme change should therefore be interpreted as both national and climate-regime dependent. A single national mean trend captures the sign of change, but it misses the way aridity, elevation, and regime context shape the distributional geometry of that change.

The compound dry-hot layer adds a third conclusion. Recent thermal restructuring is increasingly embedded in years when heat and precipitation deficit occur together. For warm-season precipitation and maximum temperature, $RP \geq 10$ dry-hot events affected 10.35 stations per year in 1991-2007 and 41.53 stations per year in 2008-2024; $RP \geq 20$ events increased from 2.59 to 16.12 stations. The warm-season connectedness signal also strengthened, with Moran's I slopes of $+0.075$ and $+0.087$ per decade for $RP \geq 10$ and $RP \geq 20$ events, respectively. Perhaps most importantly, the dominant event character shifted from dry-dominant in

the early period to hot-dominant in the late period. This does not prove causal forcing, but it does show that recent hot-season compound risk in the station archive is increasingly structured around heat excess.

These findings matter for more than descriptive trend reporting. Scientifically, they show that recent thermal change across Iran cannot be reduced to a single smooth national field or a single representative mean trend. The strongest changes emerge in the relative behavior of lower, central, and upper quantiles. In practical terms, this means that heat-risk assessment, agricultural planning, water-resource management, and climate adaptation strategies should pay particular attention to the rising frequency of exceptionally warm-day years and the simultaneous erosion of cool-day and cool-night conditions. This relevance extends beyond meteorological monitoring because recent heat-stress research shows that climate hazards translate into societal risk through exposure, physiological limits, and unequal adaptive capacity (Peng et al., 2026; Perkins-Kirkpatrick et al., 2026). The results also suggest that low-elevation and several coastal or interior-hot stations are especially exposed to strong distributional shifts, making quantile-aware monitoring relevant for regional early warning and climate-risk prioritization.

This implication aligns with recent heat-risk research showing that risk relevance is not defined by physical hazard intensity alone. Deadly-heat thresholds, heat-attributable mortality, and record-breaking extremes depend on the interaction between meteorological change, exposure, and adaptive limits (Mora et al., 2017; Vicedo-Cabrera et al., 2021; Fischer et al., 2021). In that context, the present station-based analysis provides the physical-risk layer needed to support more differentiated adaptation questions: which regions are experiencing the most asymmetric thermal restructuring, which climate regimes show the strongest signal emergence, and where should future vulnerability and governance assessments be focused.

More broadly, the study underscores a methodological point with wider relevance for climatology. In arid, topographically complex, and spatially heterogeneous environments, mean-only analyses can obscure the most consequential parts of the thermal signal. A quantile-aware, station-based framework is better suited to reveal whether change is distribution-wide, upper-tail dominated, or regionally uneven. The present framework therefore offers a transferable template for other dryland and mountainous regions where national climate-change assessments might otherwise smooth over exactly the contrasts that matter most for extremes.

The conclusions should also be read in light of the limits of the current dataset. The record length is 34 years, the main thermal thresholds are derived from the available 1991-2024 period, and the fixed-baseline sensitivity uses only 1991-2007 as its early reference. The dry-hot return periods are empirical rarity measures within this same short archive, not estimates of very rare century-scale events. The daily series were screened but not fully homogenized, Köppen-Geiger regimes simplify local climatic gradients, and some regionalization outcomes remain exploratory rather than uniquely resolved. Future work should therefore extend this analysis with metadata-supported daily homogenization, explicit spatial quantile models, reanalysis or model-based attribution comparisons, and physically richer diagnostics involving circulation, humidity, cloudiness, soil moisture, evaporative demand, and surface energy balance. Compound heat metrics, dry-hot event duration, and event-based heatwave characteristics would also help connect the distributional changes identified here to human and environmental impact pathways. This step is especially relevant given recent evidence that global heat-stress impacts are increasingly shaped by exposure inequality and physiological survivability limits (Peng et al., 2026; Perkins-Kirkpatrick et al., 2026). Even with these caveats, the central implication remains consistent: recent thermal-extreme change across Iran is asymmetric, climatically differentiated, regionally structured, hydroclimatically compounded, and best understood when tail behavior and driver concurrence are treated as primary components of climate inference rather than as secondary descriptive details.

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Author contributions

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Competing interests

`[[Insert final competing-interests declaration here, for example: "The authors declare no competing interests."]]`

Data availability

The analysis uses daily station temperature and precipitation observations for Iran over 1991-2024. `[[Insert the formal source, access conditions, and repository or request route for the raw station data.]]` Derived tables, figure outputs, and workflow summaries used to support the manuscript are organized in the project `outputs/` and `reports/` directories, with compound dry-hot outputs stored separately under `outputs/compound_dry_hot/`.

Code availability

The analysis workflow is implemented in the project scripts under `src/paper_pipeline/`, with the main execution entry point in `run_analysis.py` and configuration details in `config.yaml`. `[[Insert final public repository, archive DOI, or access statement before submission.]]`

AI tool use statement

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Supplementary material

The Supplementary Material includes the following items cited in the manuscript:

1. Supplementary Figure S1: [ijoc_data_quality_homogeneity.md](#)
2. Supplementary Figures S2-S4: [figure_5_tau_0.10.md](#), [figure_6_tau_0.50.md](#), [figure_7_tau_0.90.md](#)
3. Supplementary Figures S5-S7: [figure3_tau_0.10.md](#), [figure3_tau_0.50.md](#), [figure3_tau_0.90.md](#)
4. Supplementary Figure S8: [ijoc_homogeneity_sensitivity.md](#)
5. Supplementary Figure S9: [ijoc_bootstrap_depth_sensitivity.md](#)
6. Supplementary Figure S10: [ijoc_alternative_clustering_sensitivity.md](#)
7. Supplementary Figure S11: [cluster_spatial_validation.md](#)
8. Supplementary Figure S12: [raw_vs_fdr_counts.md](#)
9. Supplementary Figure S13: [moran_i_heatmap.md](#)

10. **Supplementary Figures S14-S16:** [fixed_baseline_period_change_summary.png](#), [warming_link_network_quantile_response.png](#), [climate_signal_emergence_q50_maps.png](#)
11. **Supplementary Figures S17-S19:** [koppen_geiger_station_regimes.png](#), [climate_regime_fixed_baseline_shifts.png](#), [climate_regime_fingerprint_scores.png](#)
12. **Supplementary Tables S1-S3:** [data_quality_homogeneity_overview.md](#), [data_quality_station_summary.md](#), [data_homogeneity_tests_station_summary.md](#)
13. **Supplementary Table S4:** [homogeneity_flag_exclusion_sensitivity.md](#)
14. **Supplementary Tables S5-S6:** [bootstrap_depth_sensitivity_summary.md](#), [bootstrap_depth_sensitivity_station_comparison.md](#)
15. **Supplementary Tables S7-S8:** [alternative_clustering_sensitivity_summary.md](#), [alternative_clustering_assignments.md](#)
16. **Supplementary Table S9:** [regional_cluster_spatial_validation.md](#)
17. **Supplementary Table S10:** [station_significance_fdr.md](#)
18. **Supplementary Table S11:** [spatial_autocorrelation_moran.md](#)
19. **Supplementary Tables S12-S18:** [regional_temperature_anomaly.csv](#), [fixed_baseline_period_change_summary.csv](#), [warming_link_network_quantile_response.csv](#), [climate_signal_emergence_summary.csv](#), [climate_fingerprint_component_scores.csv](#), [climate_fingerprint_network_components.csv](#), [climate_fingerprint_permutation_null.csv](#)
20. **Supplementary Tables S19-S27:** [koppen_geiger_station_assignments.csv](#), [koppen_geiger_regime_summary.csv](#), [climate_regime_quantile_summary.csv](#), [climate_regime_fixed_baseline_summary.csv](#), [climate_regime_warming_response_summary.csv](#), [climate_regime_emergence_summary.csv](#), [climate_regime_fingerprint_components.csv](#), [climate_regime_fingerprint_summary.csv](#), [climate_regime_difference_tests.csv](#)
21. **Supplementary Figures S28-S32 and Supplementary Tables S28-S35:** [compound_dry_hot_extent_timeseries.png](#), [compound_dry_hot_station_frequency_maps.png](#), [compound_dry_hot_frequency_change_maps.png](#), [compound_dry_hot_driver_shift.png](#), [compound_dry_hot_moran_connectedness.png](#), [compound_dry_hot_station_year.csv](#), [compound_dry_hot_yearly_extent.csv](#), [compound_dry_hot_trend_summary.csv](#), [compound_dry_hot_distribution_shift_tests.csv](#), [compound_dry_hot_station_frequency.csv](#), [compound_dry_hot_driver_summary.csv](#), [compound_dry_hot_moran_yearly.csv](#), [compound_dry_hot_moran_trend_summary.csv](#)

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